

34721/34722 Linear Control Design 1

Spring Semester

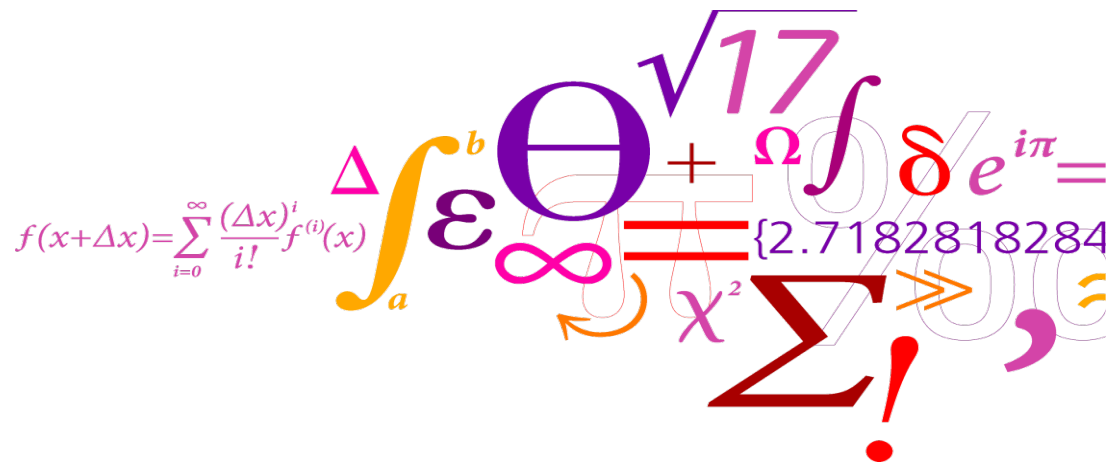
Silvia Tolu - Dimitrios Papageorgiou



Lecture 11/13

Schedule (13:00-15:00)

- **Limited systems**



What you will learn today

After this lecture you will be able to

- Explain rate and amplitude limitations in systems
- Use mitigation strategies in the control design of limited systems

Rate-limited systems

```
syms s;  
G = 900/((s + 1)*(s^2 + 50*s + 900));  
Y = vpa(simplify(ilaplace(G)),3) %  
Calculate the impulse response
```

Rate limitation: A system cannot operate arbitrarily fast because of

- **Slow (dominant) poles:** e.g. it takes few seconds to move a very big and heavy track (or a ship).
 - Dominant poles are the poles with the smallest (in absolute value) real part.

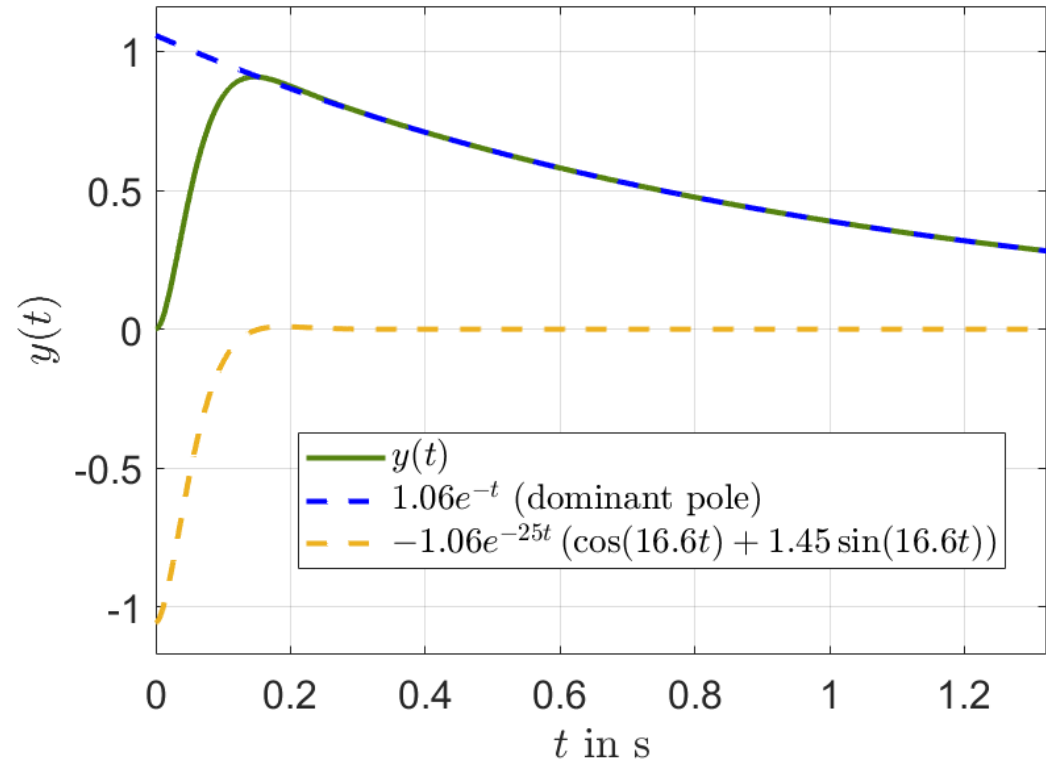
$$G(s) = \frac{900}{(s + 1)(s^2 + 50s + 900)}$$

-1 -25 ± 16.5j

- **Artificial rate limitations on the actuators:** e.g. A motor in a robotic joint has a limit in how fast it reacts to prevent wear of the mechanical parts.

$$y(t) = 1.06e^{-t} - 1.06e^{-25t}(\cos(16.6t) + 1.45\sin(16.6t))$$

Impulse response

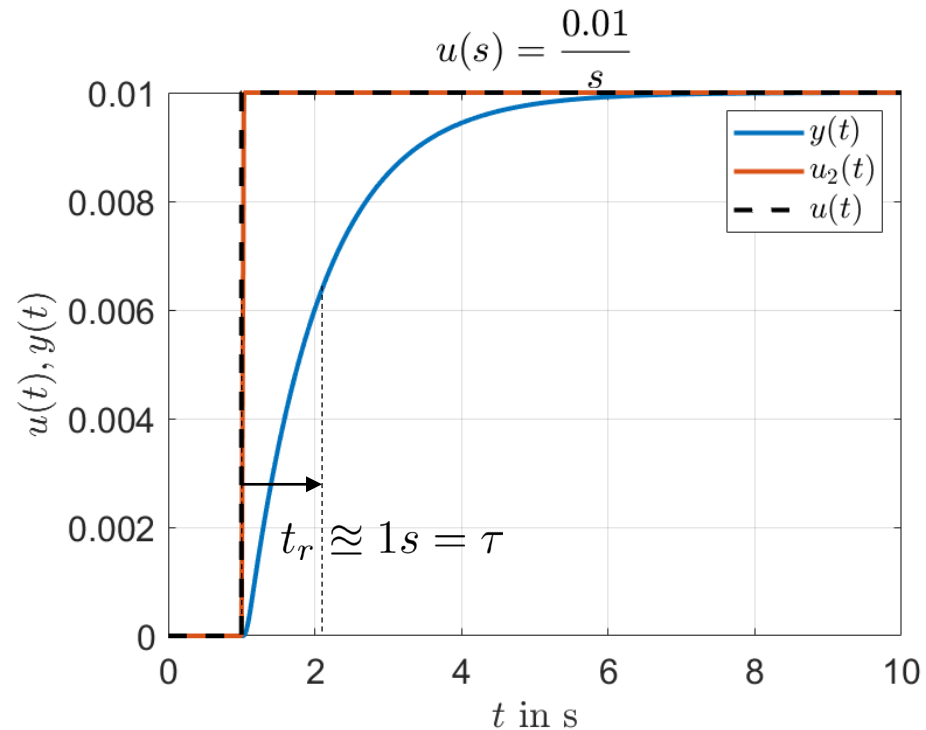
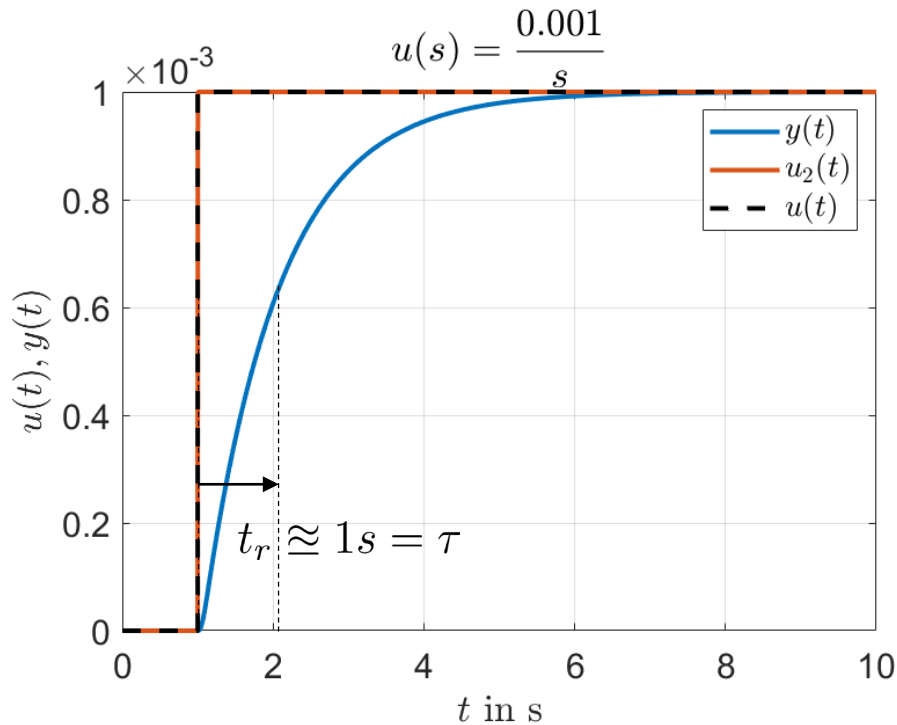
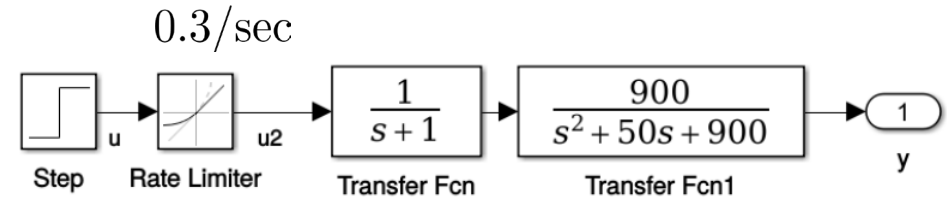


Rate-limited systems

The input signal cannot grow by more than 0.3 every second

Consider the system $G(s) = \frac{900}{(s+1)(s^2+50s+900)}$

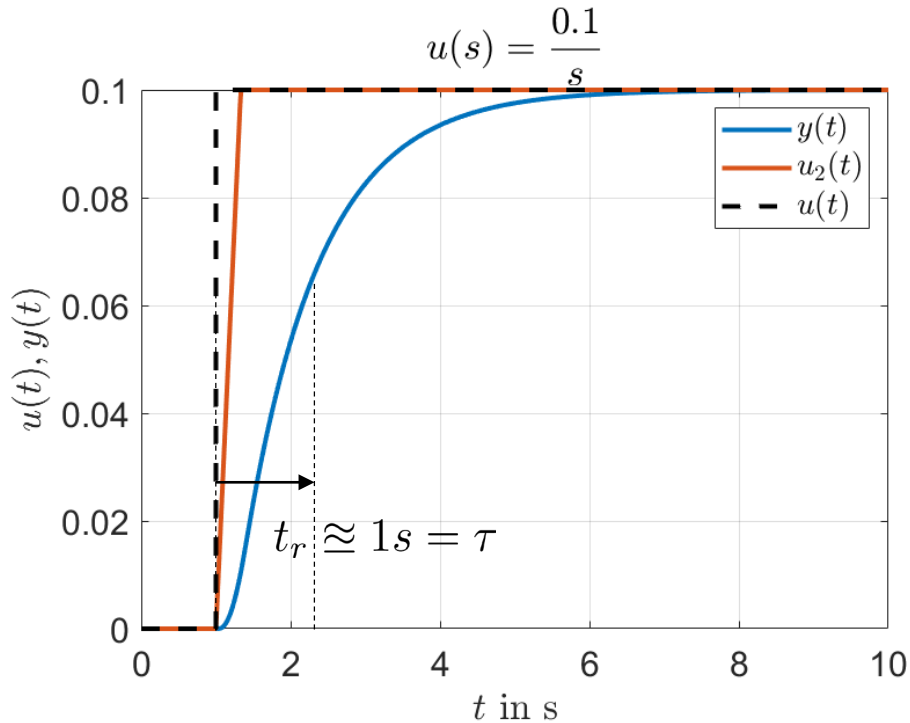
with an input rate limiter of 0.3/s .



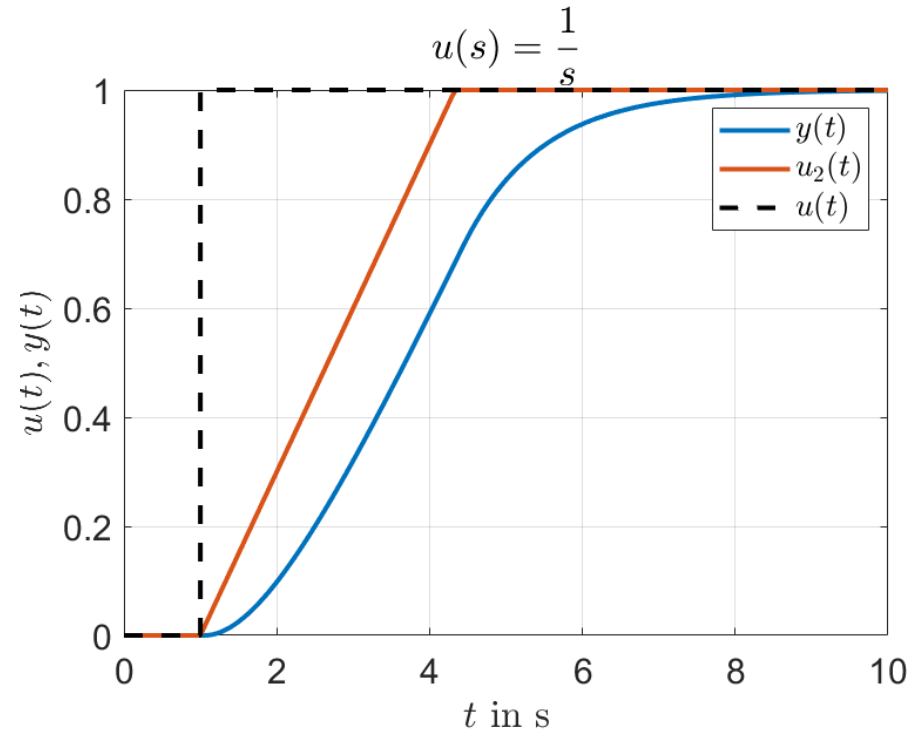
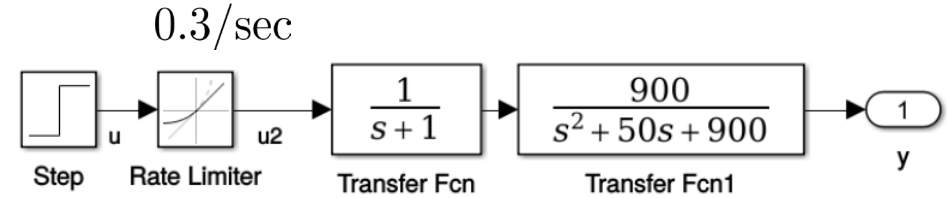
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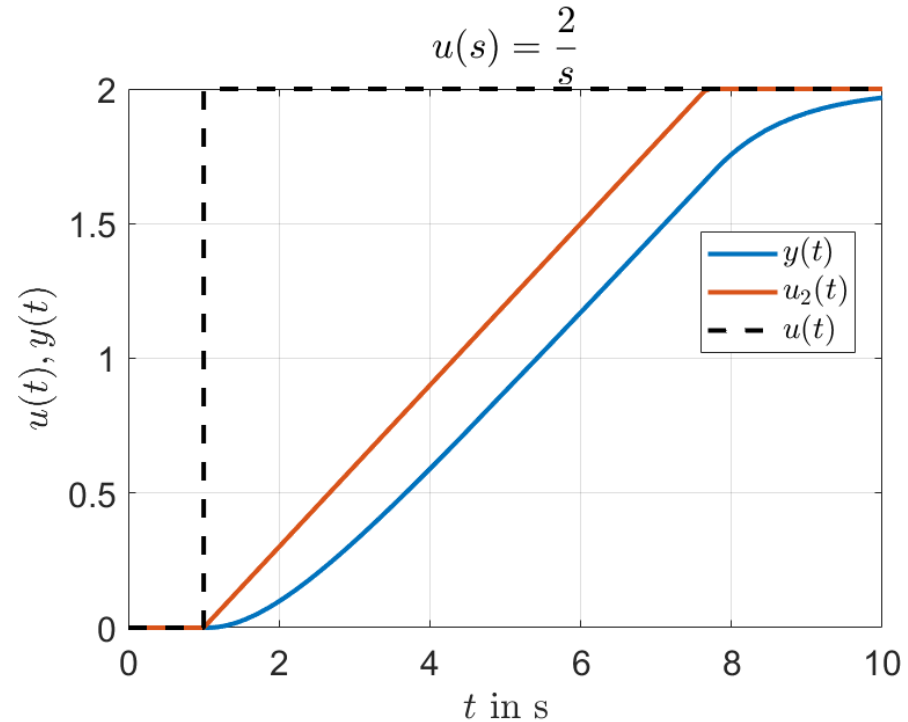
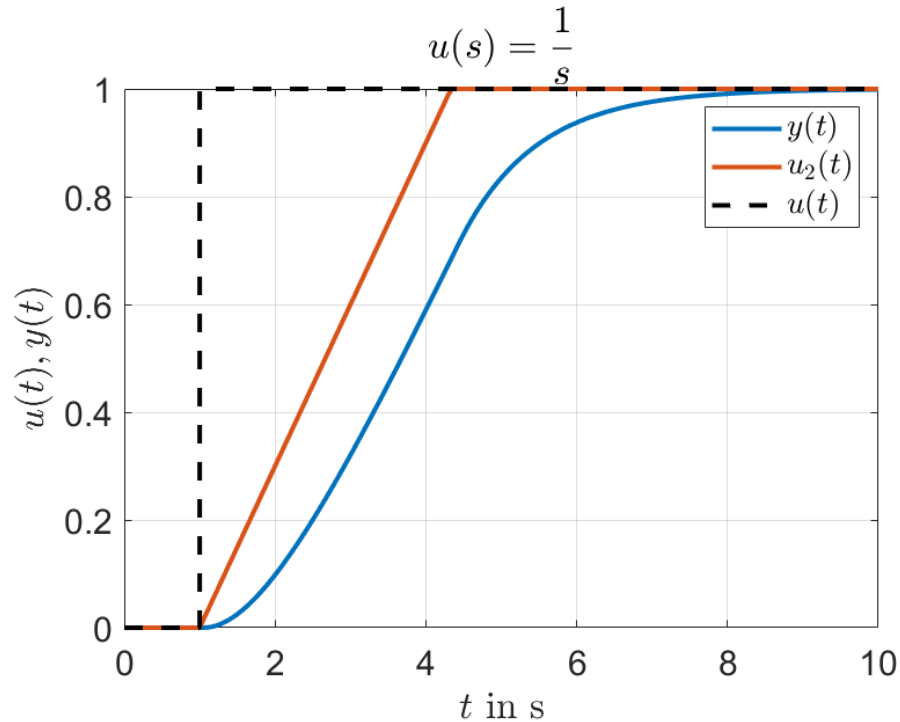
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Rate-limited systems

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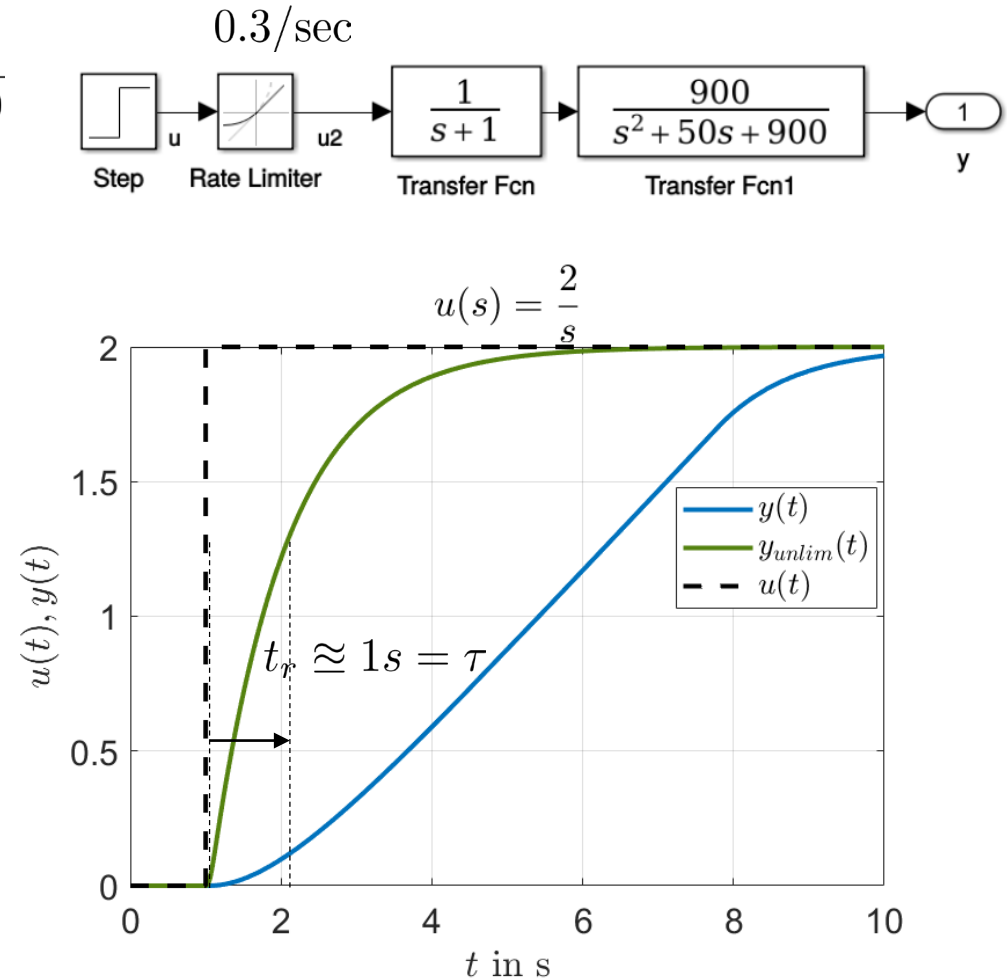
Rate-limited systems

The input signal cannot grow by more than 0.3 every second

Consider the system $G(s) = \frac{900}{(s+1)(s^2+50s+900)}$

with an input rate limiter of 0.3/s .

- For small steps the system behaves as expected.
- Estimation of transfer function from step response is reliable only for small steps.
- If u is the output of a controller designed for the system without the rate limiter, the results may be very different than expected.



Rate-limited systems

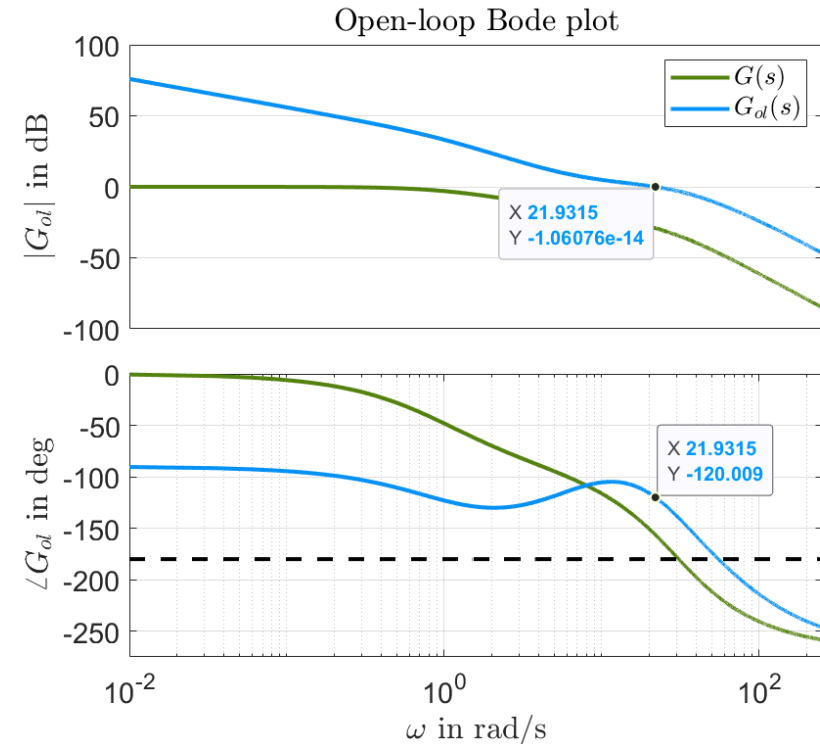
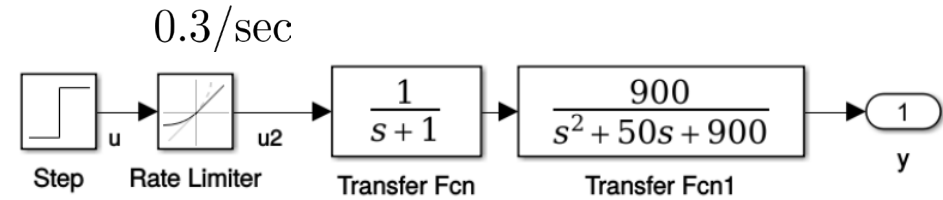
Consider the system $G(s) = \frac{900}{(s+1)(s^2+50s+900)}$

with an input rate limiter of 0.3/s . Design a PI-Lead controller:

- Choose $N_i = 3$, $\alpha = 0.1$, $\gamma_M = 60^\circ = 1.0472 \text{ rad}$.
- $-\pi + \gamma_M = \angle G(j\omega_c) - \arctan\left(\frac{1}{3}\right) + \arcsin\left(\frac{1-0.1}{1+0.1}\right) \Leftrightarrow$
- $\Rightarrow \omega_c \cong 22 \text{ rad/s} \Rightarrow \tau_i = \frac{N_i}{\omega_c} = 0.137$, $\tau_d = \frac{1}{\omega_c \sqrt{\alpha}} = 0.144$

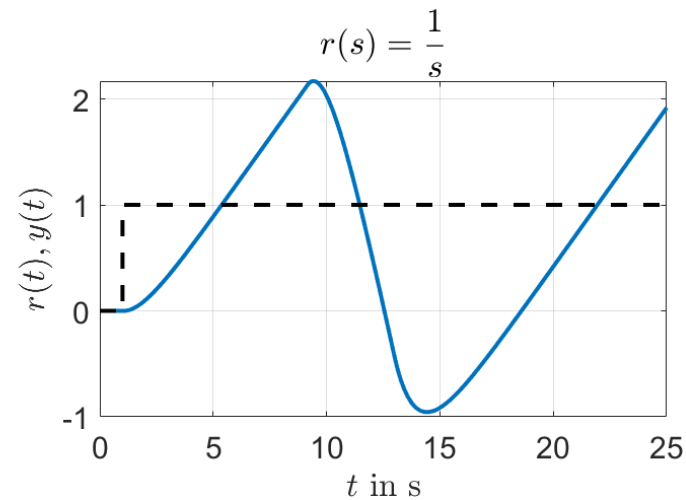
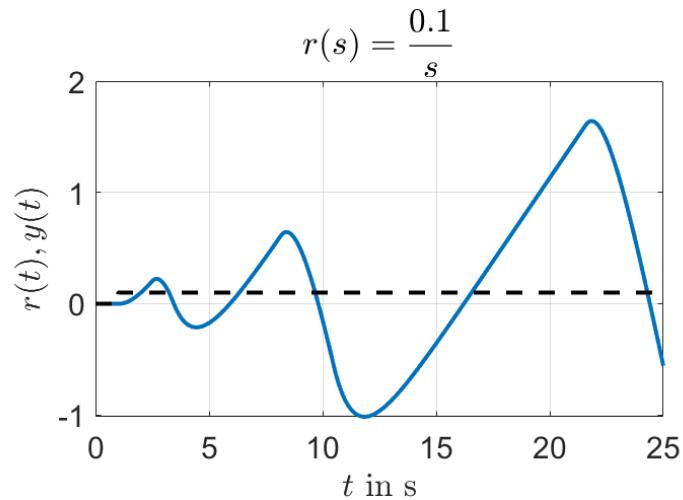
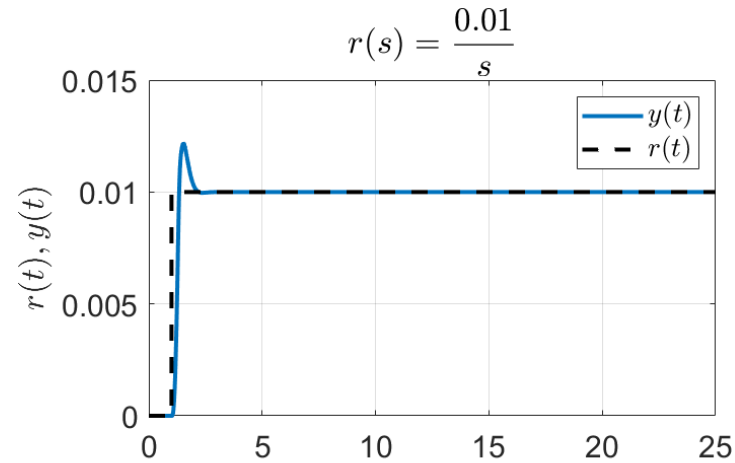
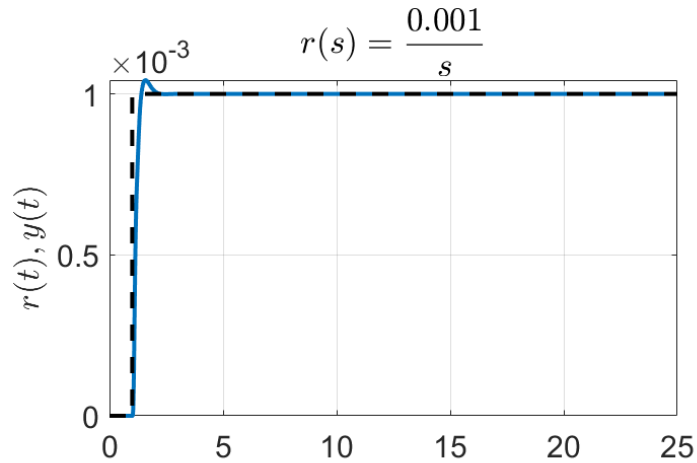
$$K_P = \frac{1}{|C_{PI}(s)C_D(s)G(s)|} \Big|_{s=j\omega_c} = 8.6$$

The input signal cannot grow by more than 0.3 every second



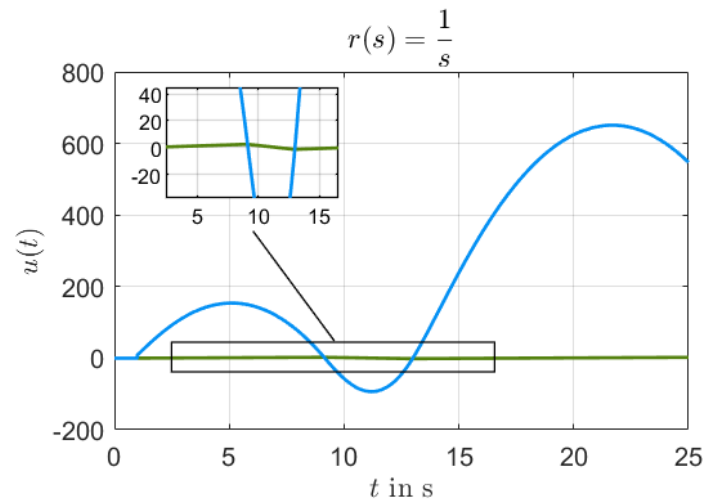
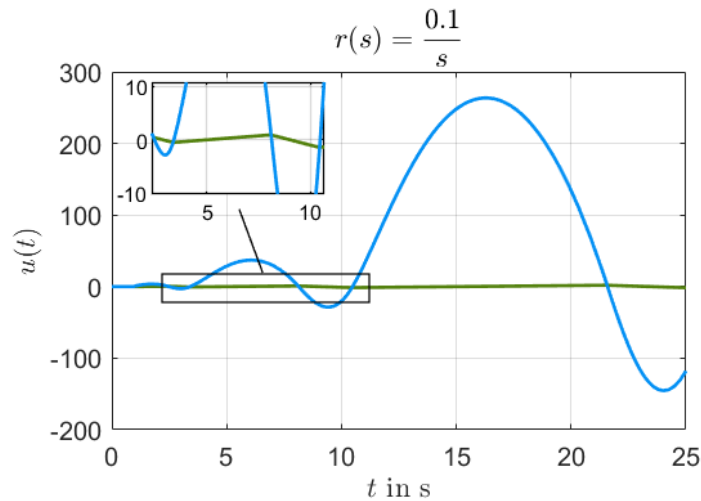
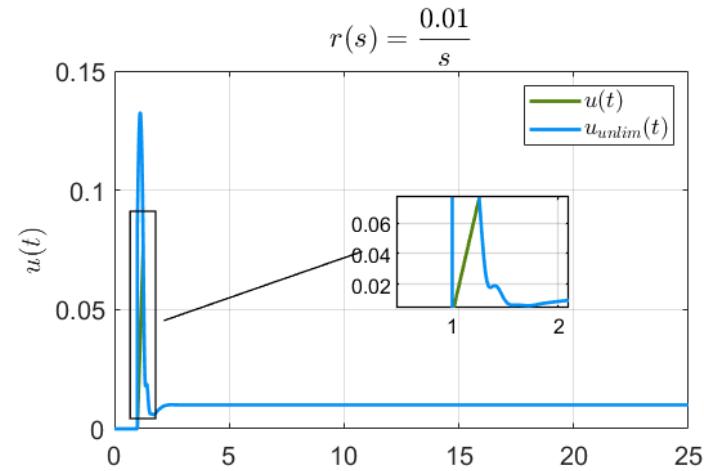
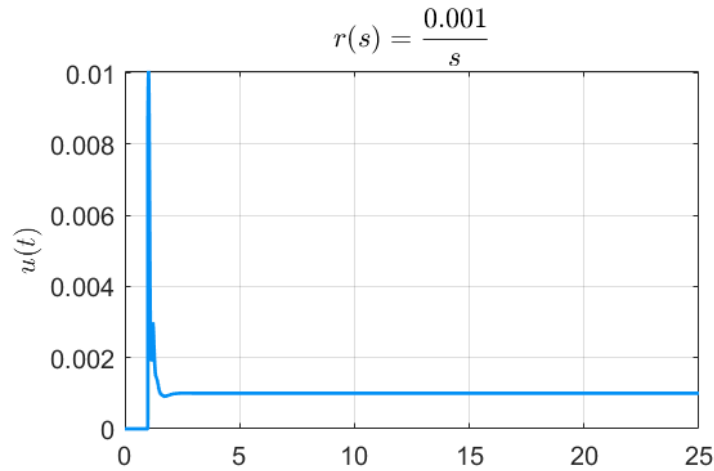
Rate-limited systems

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Rate-limited systems

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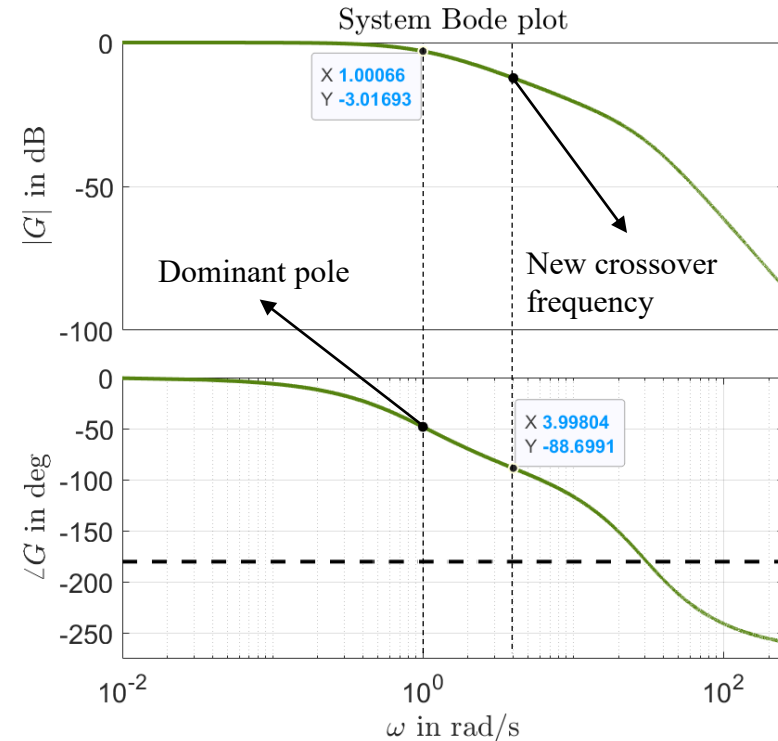
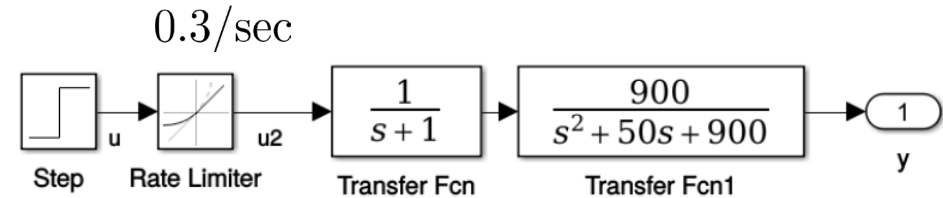
Rate-limited systems

Consider the system $G(s) = \frac{900}{(s+1)(s^2+50s+900)}$

with an input rate limiter of 0.3/s. Design a PI-Lead controller. **HOW TO DEAL WITH THAT?**

- Find the dominant pole of the system.
 - $s_{dp} = -1 \rightarrow \omega_{dp} = 1 \text{ rad/s}$
- Choose a **new crossover frequency**.
 - Higher than the dominant pole but not too large.
Here, $\omega_c = 4 \text{ rad/s}$
- Redesign the controller.
- Evaluate the closed-loop performance.
- Iterate the design if necessary.

The input signal cannot grow by more than 0.3 every second



Rate-limited systems

Consider the system $G(s) = \frac{900}{(s+1)(s^2+50s+900)}$

with an input rate limiter of 0.3/s. Design a PI-Lead controller. **HOW TO DEAL WITH THAT?**

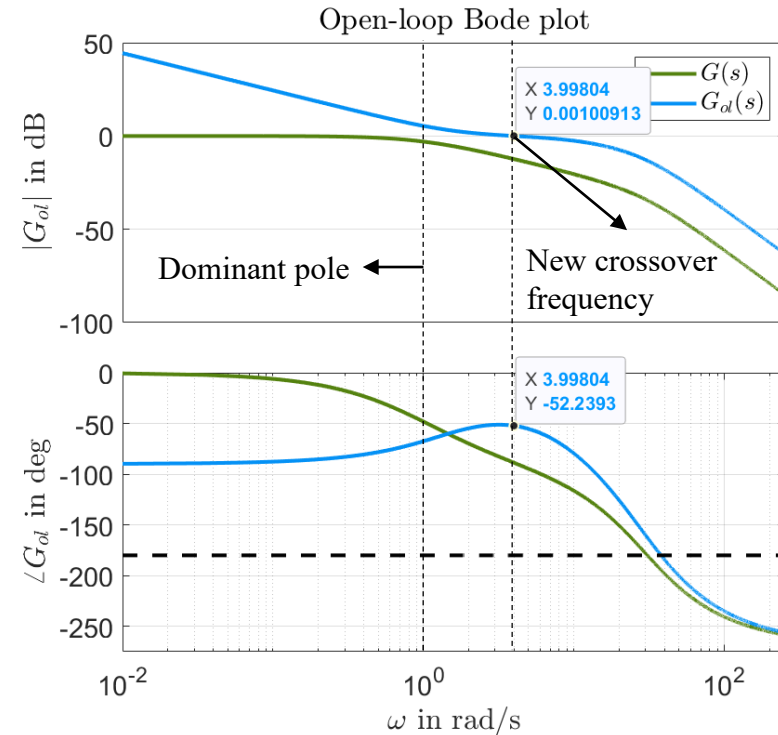
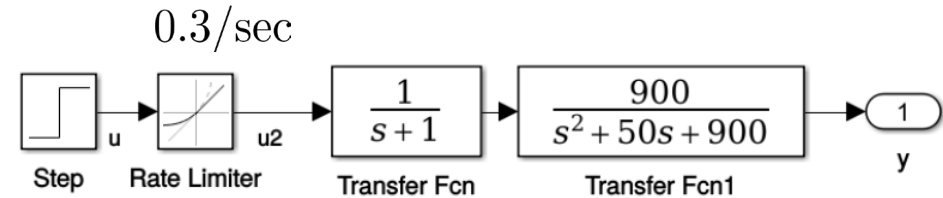
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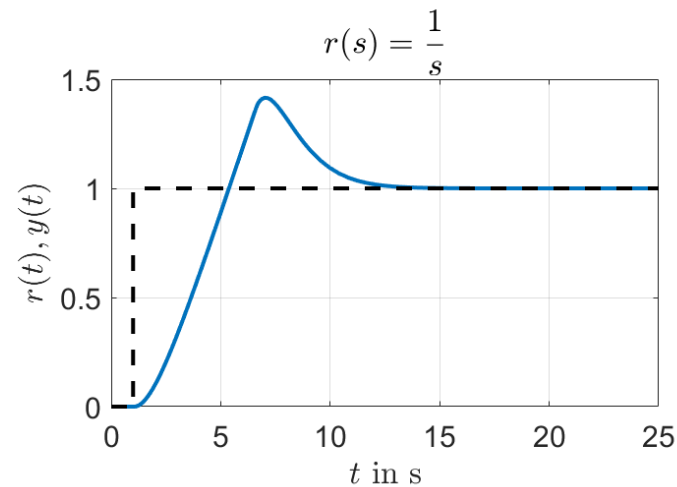
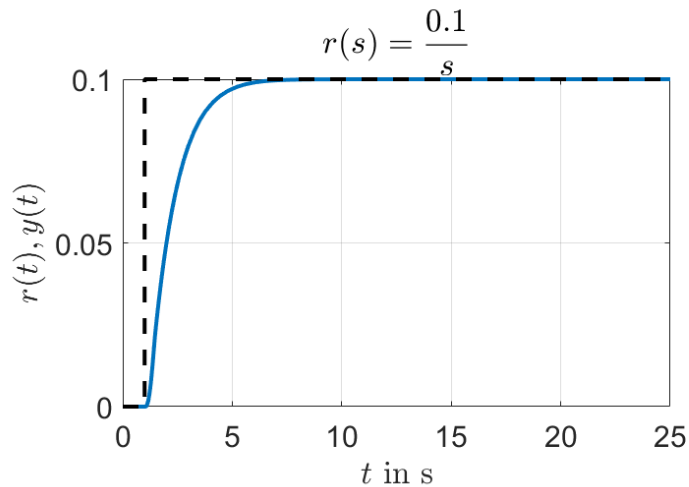
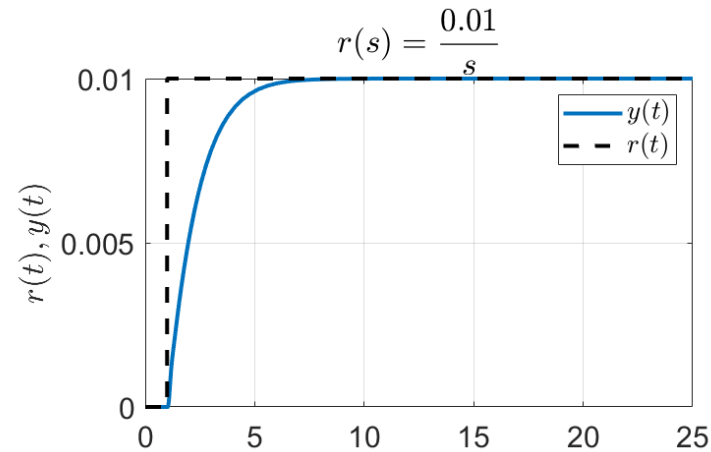
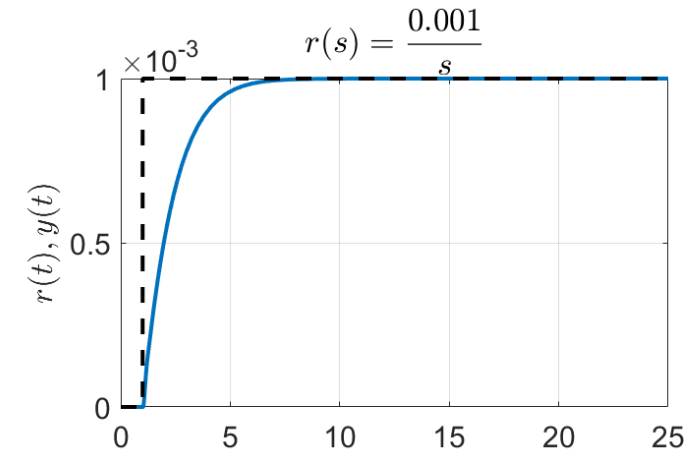
$$\tau_i = \frac{N_i}{\omega_c} = 0.75, \tau_d = \frac{1}{\omega_c \sqrt{\alpha}} = 0.79, K_P = 1.2456$$

The input signal cannot grow by more than 0.3 every second



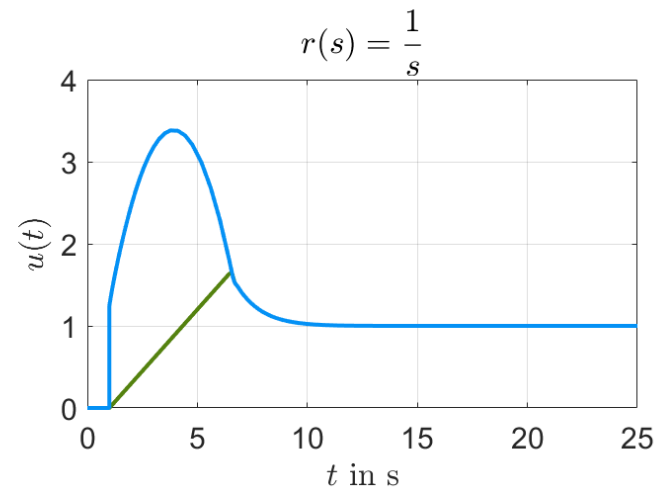
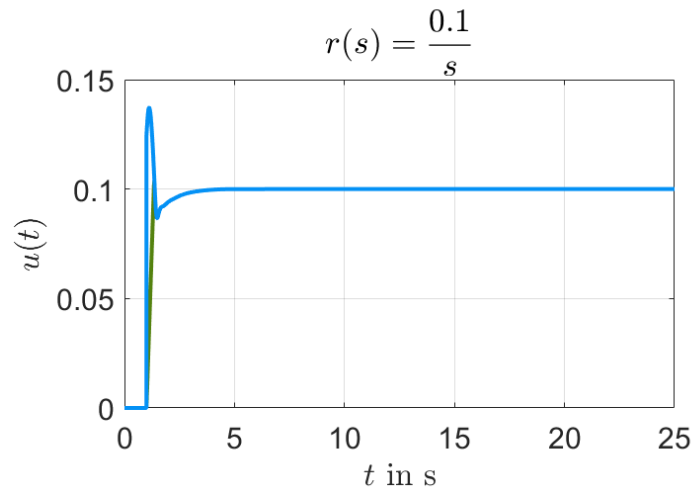
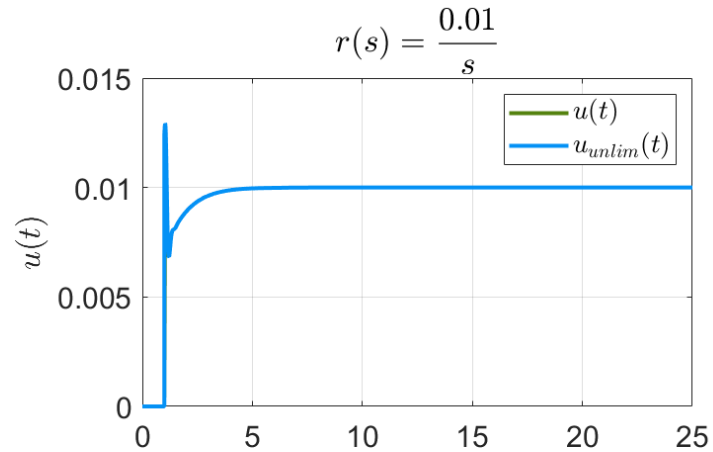
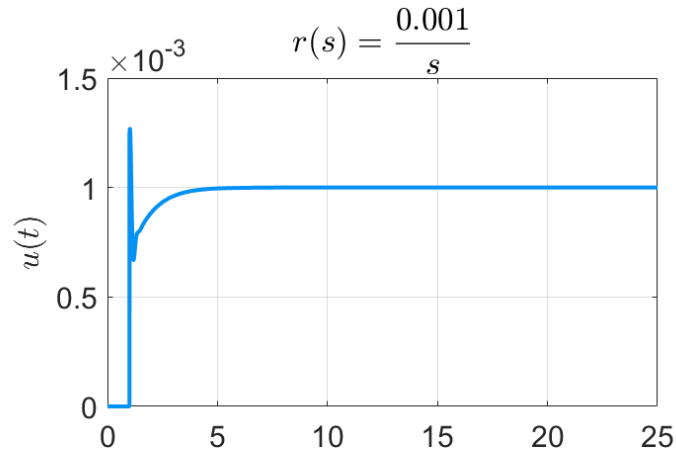
Rate-limited systems

The input signal cannot grow by more than 0.3 every second



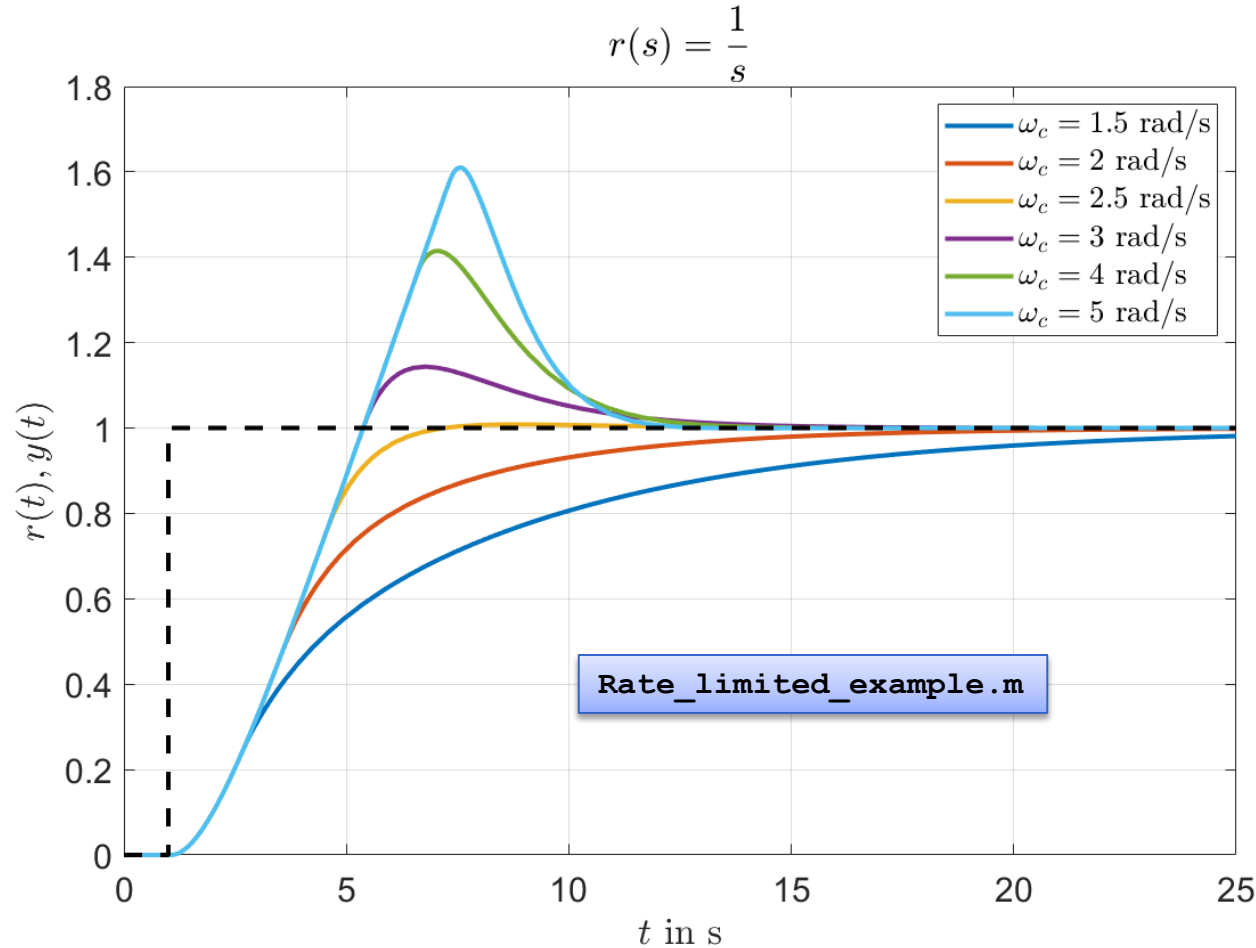
Rate-limited systems

The input signal cannot grow by more than 0.3 every second



Rate-limited systems

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Rate-limited systems

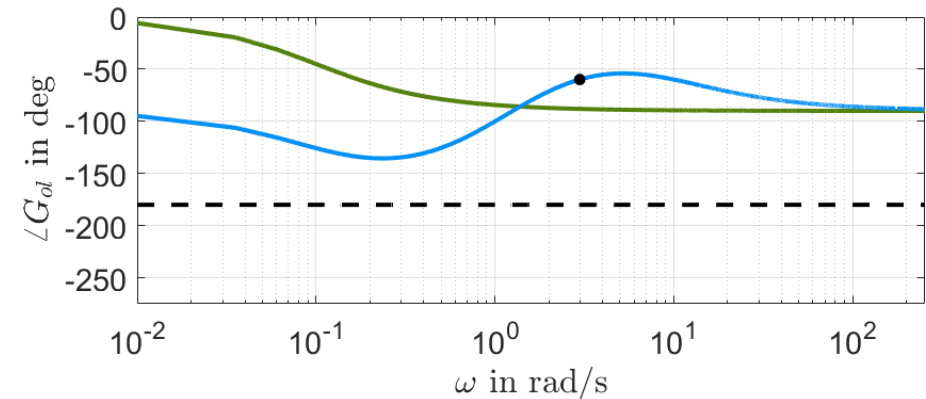
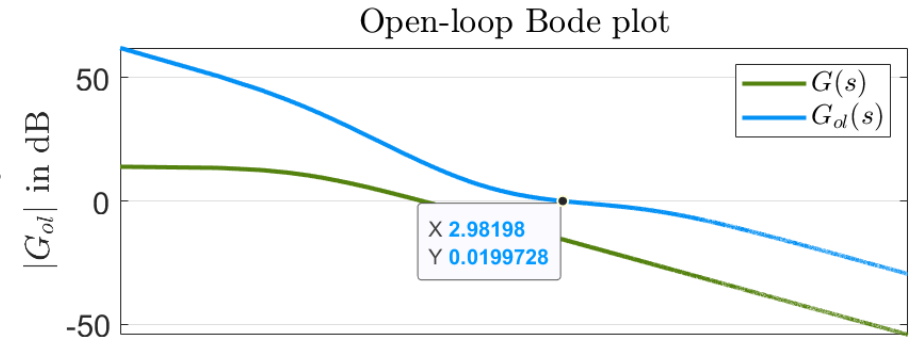
Example: $G(s) = \frac{5}{10s + 1}$

The system is rate-limited by **1/sec**. Design a PI-Lead controller ($N_i = 2$, $\alpha = 0.1$) such that the closed-loop system has rise time $t_r \leq 10$ s for small reference signals ($r(t) = 0.1$) and is stable for larger ones ($r(t) = 1$).

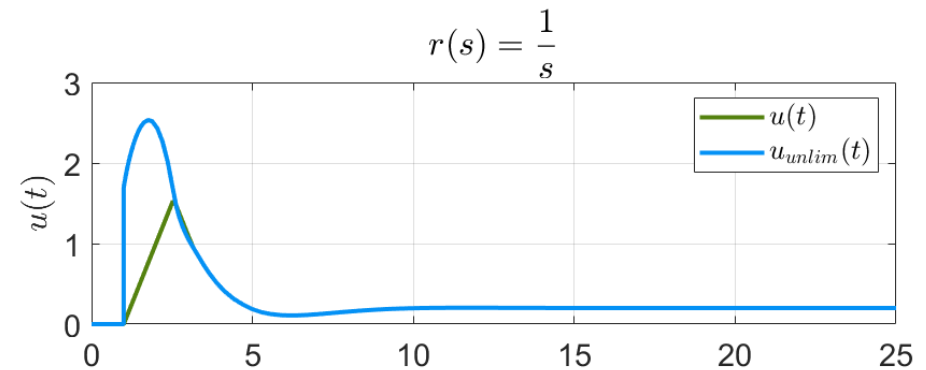
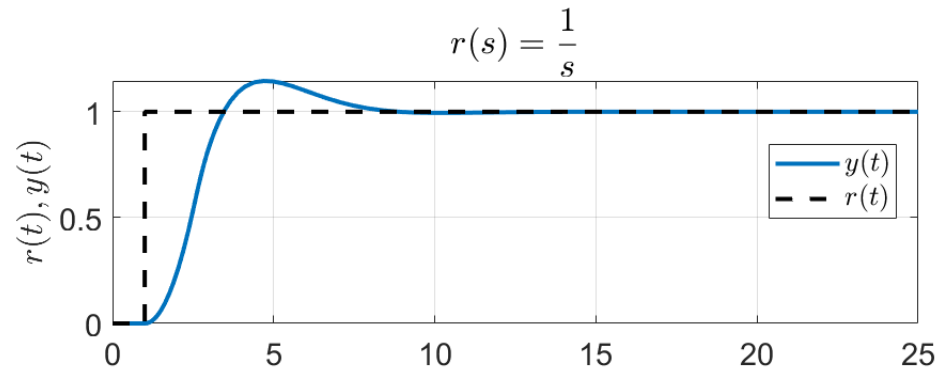
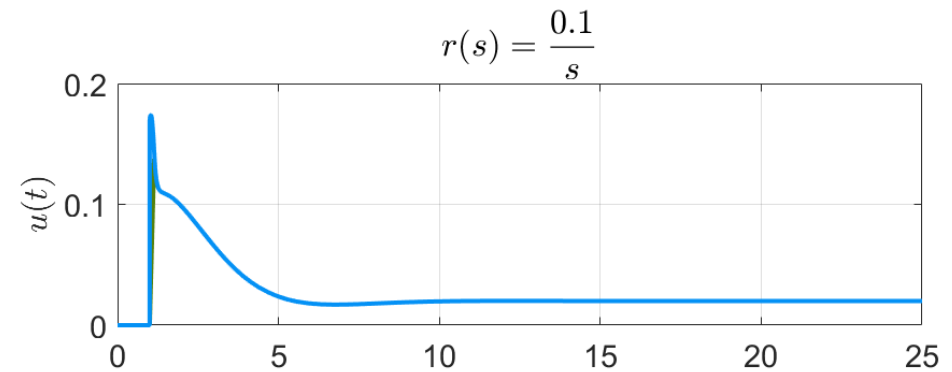
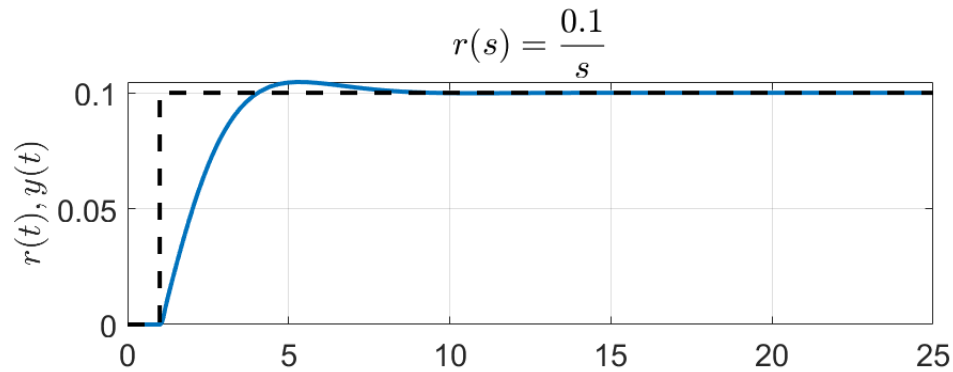
- The only pole is at $\omega = 1/\tau = 0.1$ rad/s.
- The new crossover frequency will be chosen to be larger but not too large: $\omega_c = 3$ rad/s.

- $\tau_i = \frac{N_i}{\omega_c} = 0.67$, $\tau_d = \frac{1}{\omega_c \sqrt{\alpha}} = 1.054$, $K_P = 1.7$

- $$G_{ol}(s) = \underbrace{1.7 \frac{0.67s + 1}{0.67s}}_{\text{PI}} \cdot \underbrace{\frac{1.054s + 1}{0.1504s + 1}}_{\text{Lead}} \cdot \underbrace{\frac{5}{10s + 1}}_{G(s)}$$



Rate-limited systems



Rate-limited systems

Comprehension questions

- Is a rate-limited system nonlinear?
 - Yes and this causes our initial design to possibly fail.
- How can we find the transfer function of a rate-limited system?
 - From laws of physics plus linearisation (white box modelling).
 - From measurements (black box identification) but we need to use small inputs such that the rate limitation is not activated.
- Can a rate-limited system with a stable controller design become unstable?
 - Yes and this is due to not taking into account the nonlinearity of the rate limiter.
- How can we design a good controller when we have a rate-limited system?
 - Choose a crossover frequency higher than the dominant pole (but not too larger).
 - Follow the usual steps for control design (e.g. for a PI-Lead controller).
 - Evaluate the step response of the **real** closed-loop system (with the rate limiter on).
 - Iterate the design (select different crossover frequencies) if the results are not satisfactory.

Amplitude limitations on the system's input

Consider the system in the block diagram with an input saturation (amplitude limitation) at ± 4 (e.g. 4 V for a motor), where

$$G(s) = \frac{178}{(s+1)(0.003s+1)}, \quad H(s) = \frac{1}{0.0009s+1}$$

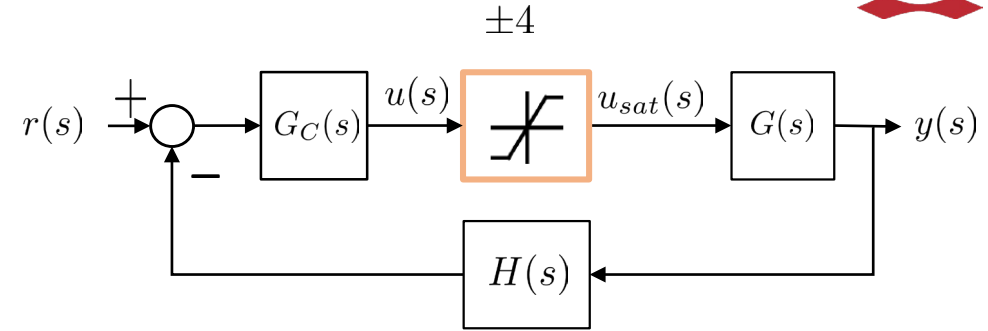
We want to control the system with a PI-Lead controller such that we have zero steady-state error and 70° phase margin.

- With $N_i = 5$ and $\alpha = 0.1$ we have

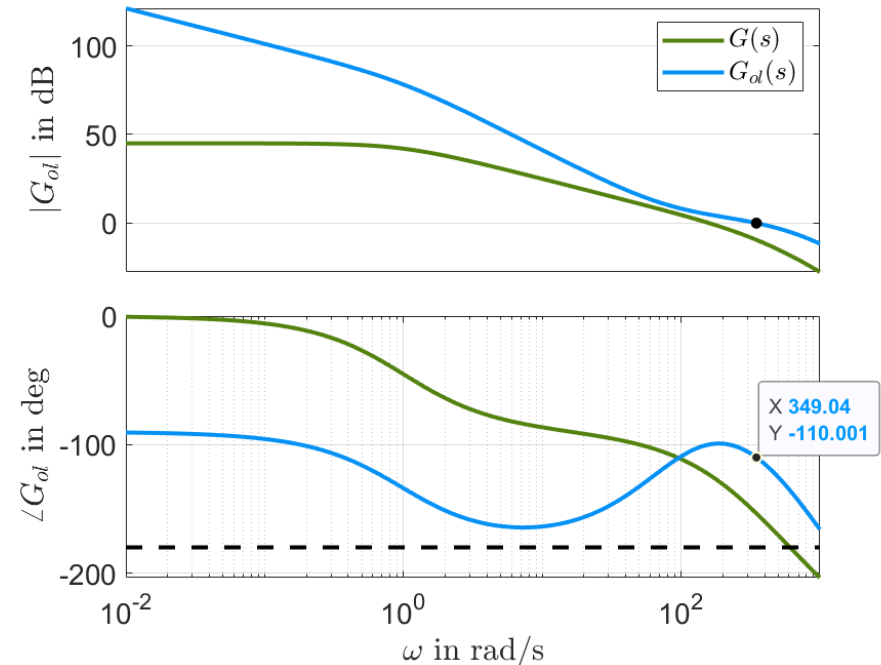
$$-\pi + \gamma_M = \angle G(j\omega_c) + \arctan\left(\frac{1}{5}\right) + \arcsin\left(\frac{1-0.1}{1+0.1}\right) \Leftrightarrow$$

$$\Rightarrow \omega_c \cong 350 \text{ rad/s} \Rightarrow \tau_i = \frac{N_i}{\omega_c} = 0.14, \quad \tau_d = \frac{1}{\omega_c \sqrt{\alpha}} = 0.0091$$

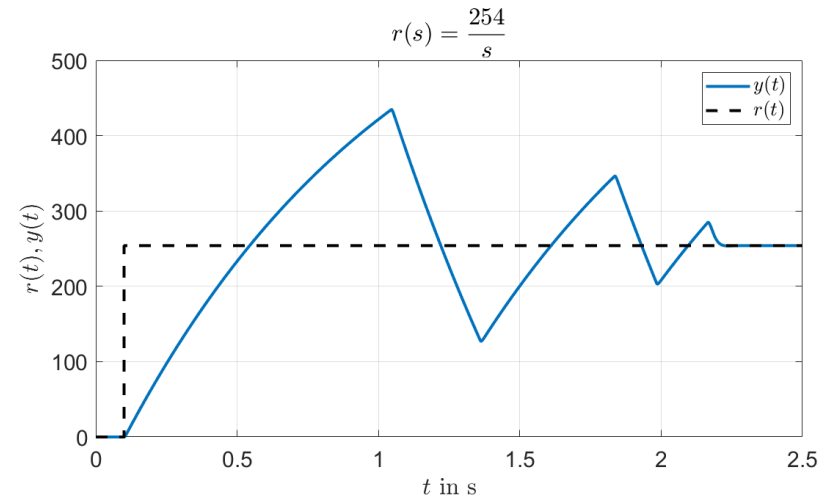
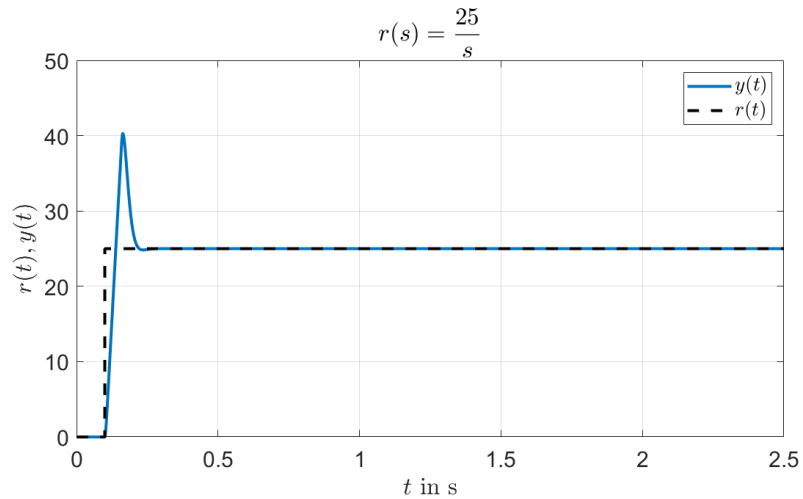
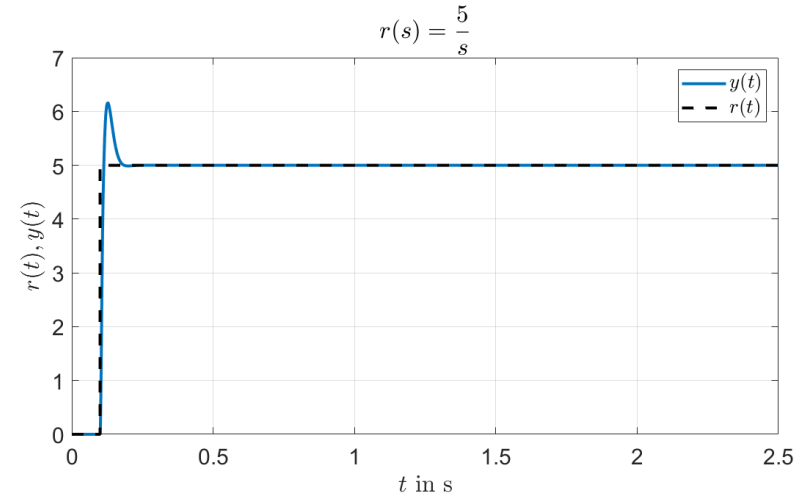
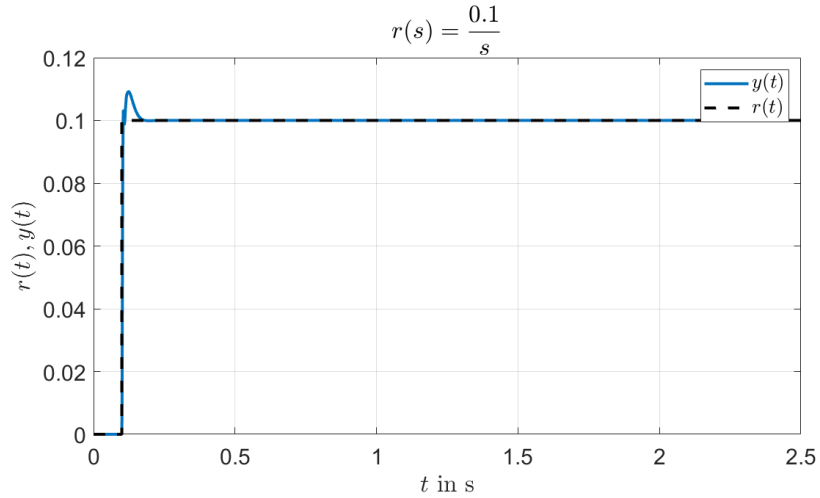
$$K_P = \left. \frac{1}{|C_{PI}(s)C_D(s)G(s)|} \right|_{s=j\omega_c} = 0.92$$



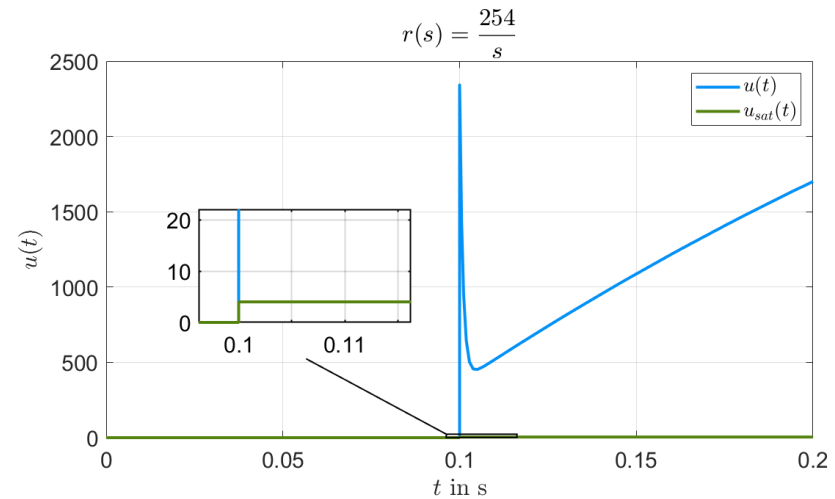
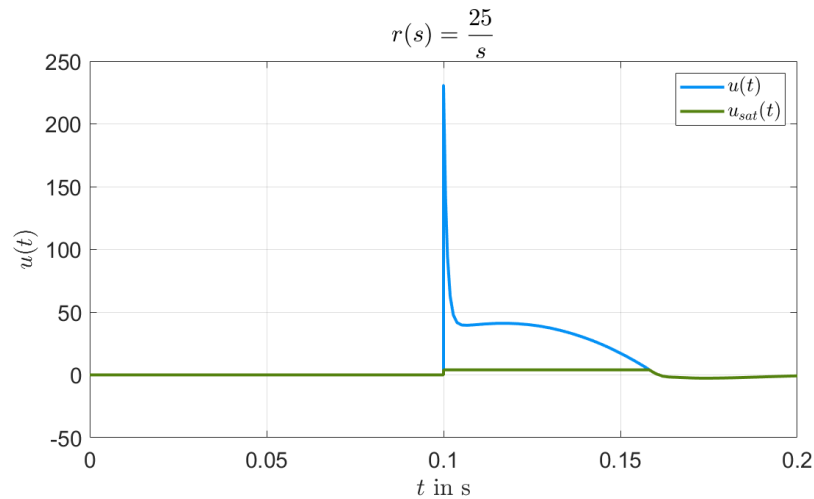
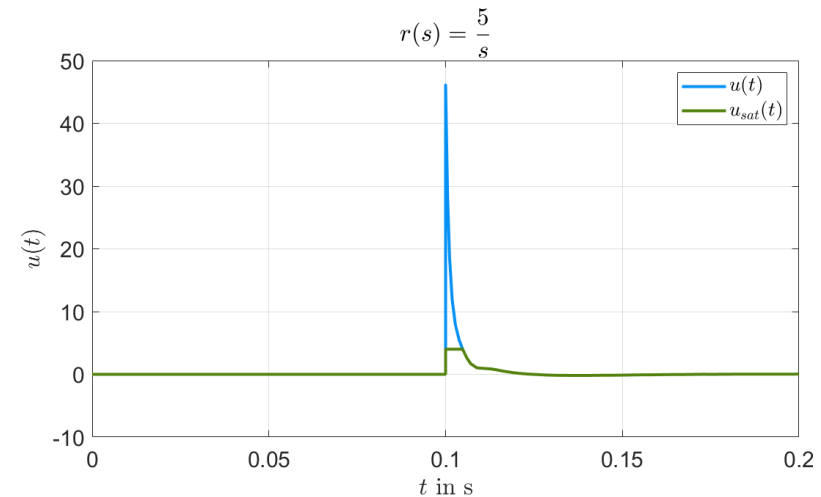
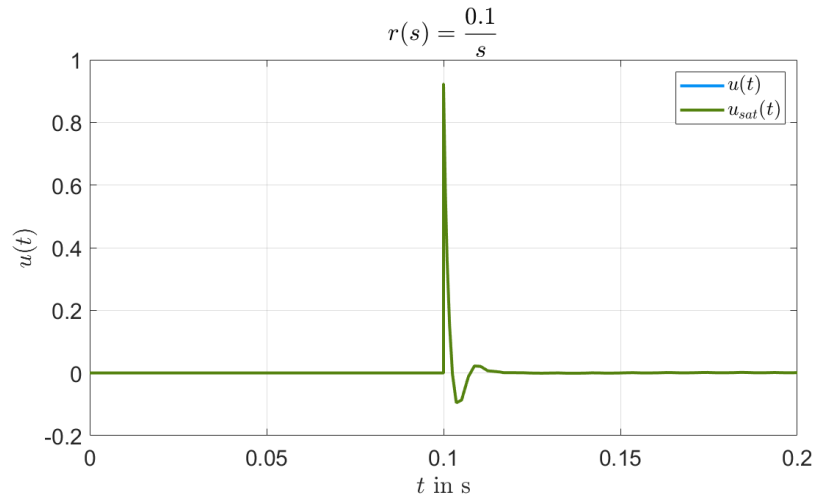
Open-loop Bode plot



Amplitude limitations on the system's input



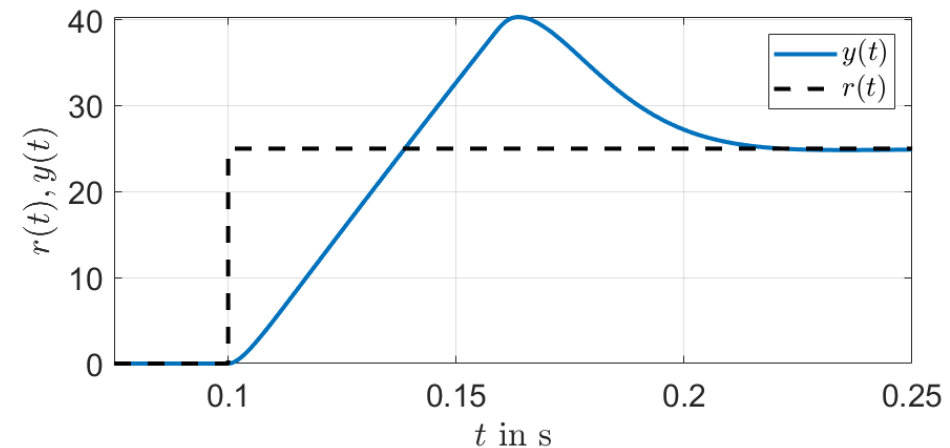
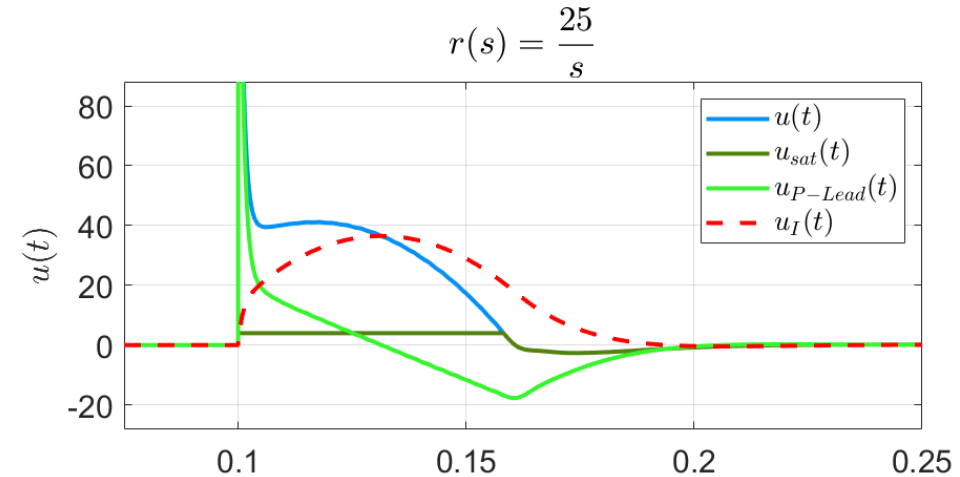
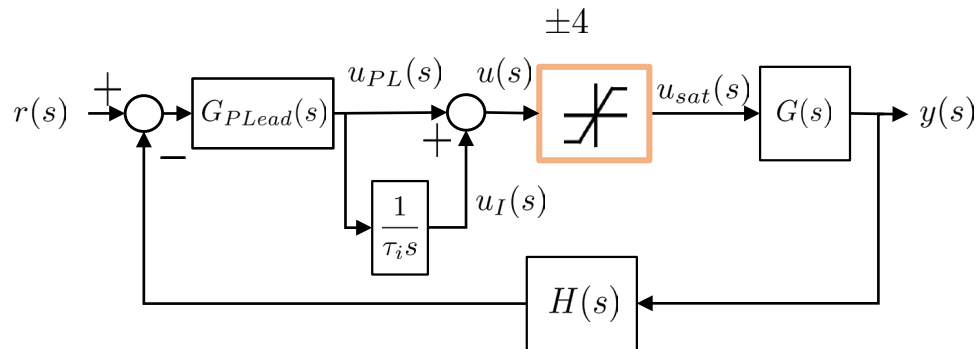
Amplitude limitations on the system's input



Amplitude limitations on the system's input

Integrator wind-up

- Once the input saturates, the integral of the error keeps increasing and so the integrator keeps “building up”.
- When the error decreases (changes sign), the large integral value needs time to decrease → so the response is delayed.

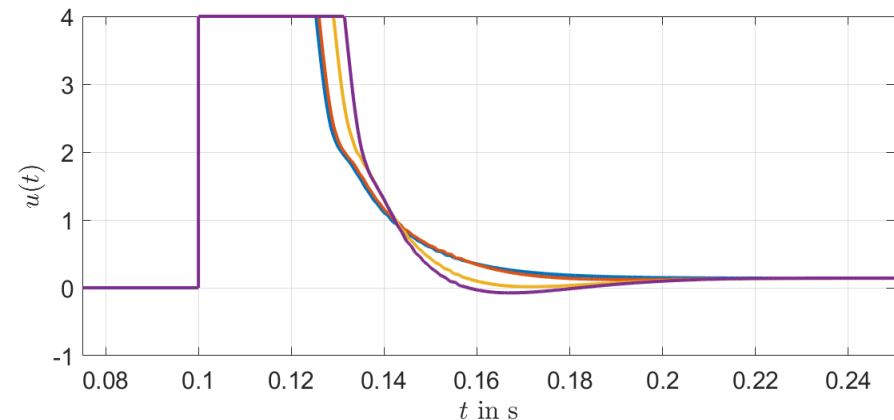
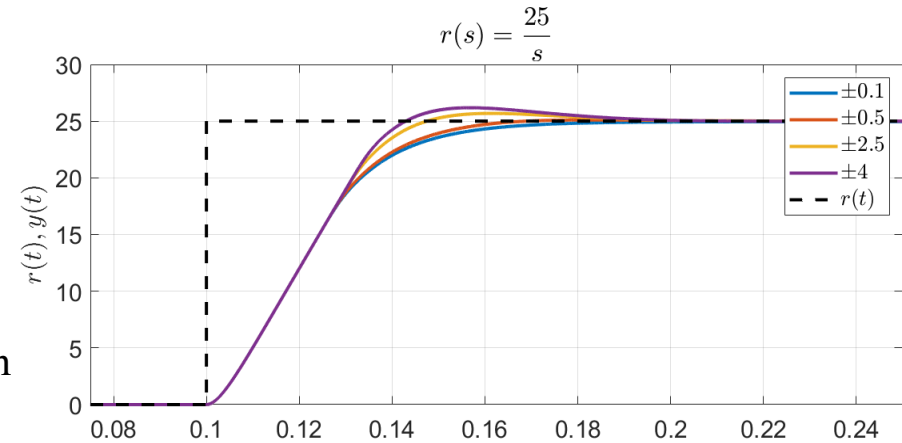
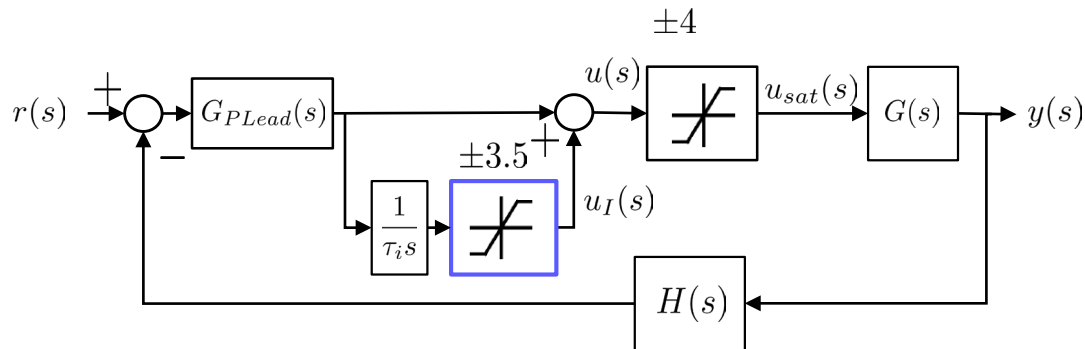


Amplitude limitations on the system's input

Integrator wind-up mitigation strategies

1. Limit the output of the integrator: Introduce a saturation block after the integrator of the PI part.

Since it is the integrator part that causes the windup effect, the limit is chosen to be equal or little smaller than the input saturation.



Amplitude limitations on the system's input

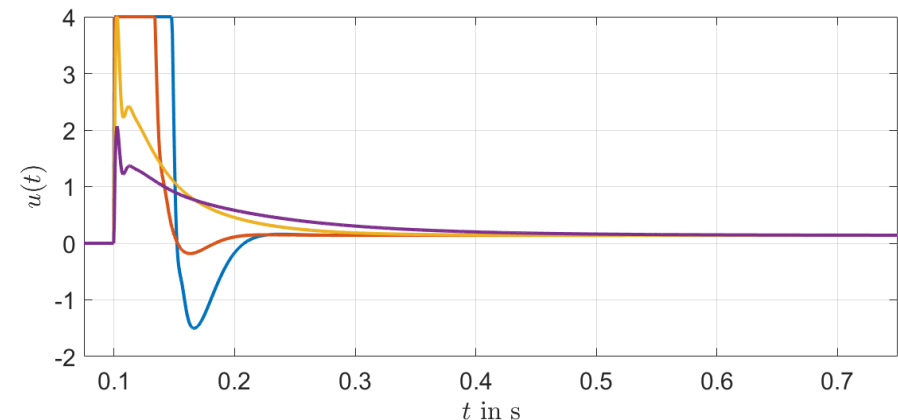
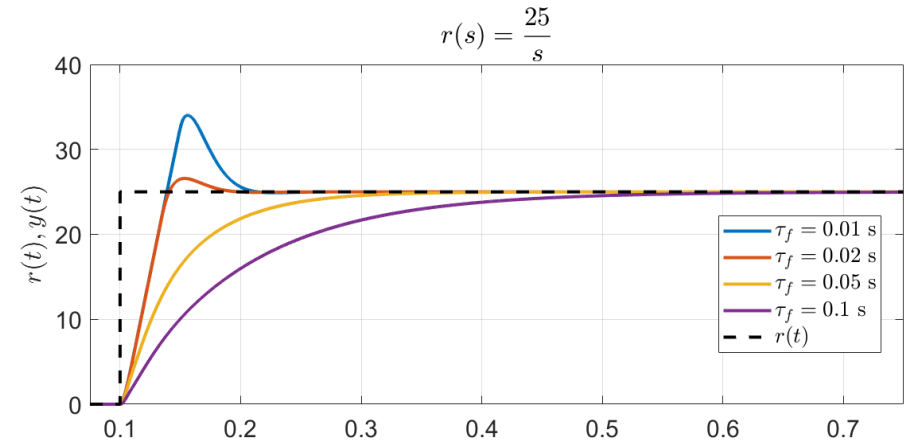
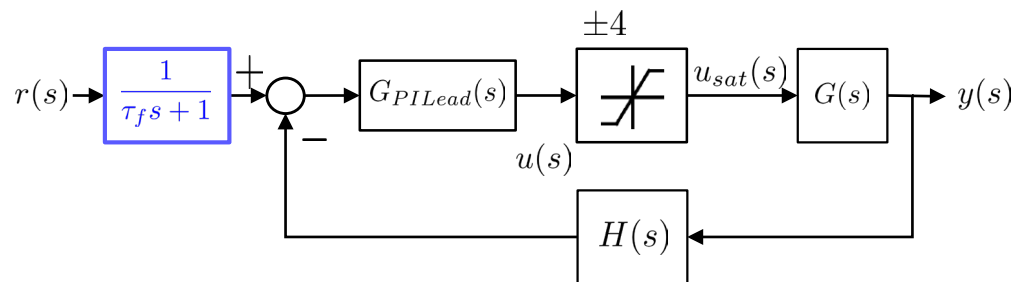
Integrator wind-up mitigation strategies

2. Insert a reference pre-filter: Introduce a filter after the reference to make it less “demanding” in control action.

Pre-filter: $G_{filt}(s) = \frac{1}{\tau_f s + 1}$

The time constant τ_f is usually chosen close to the desired rise time.

The response of the closed-loop system becomes slower even for small signals.



Amplitude limitations on the system's input

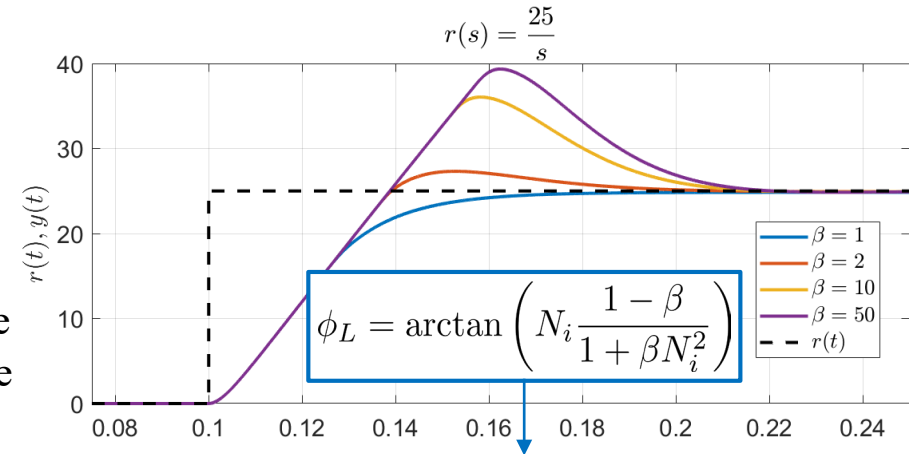
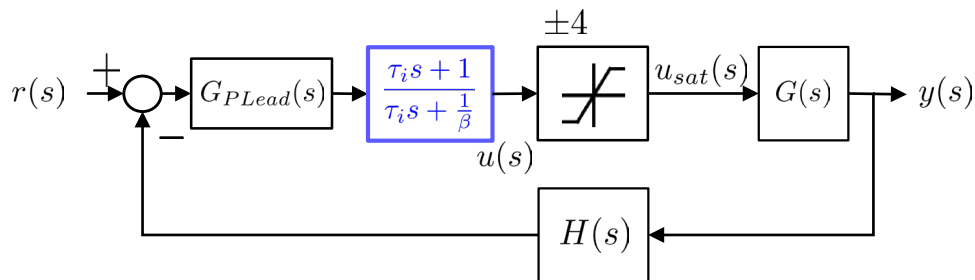
Integrator wind-up mitigation strategies

3. Lag part instead of PI (P-Lead-Lag): Replace

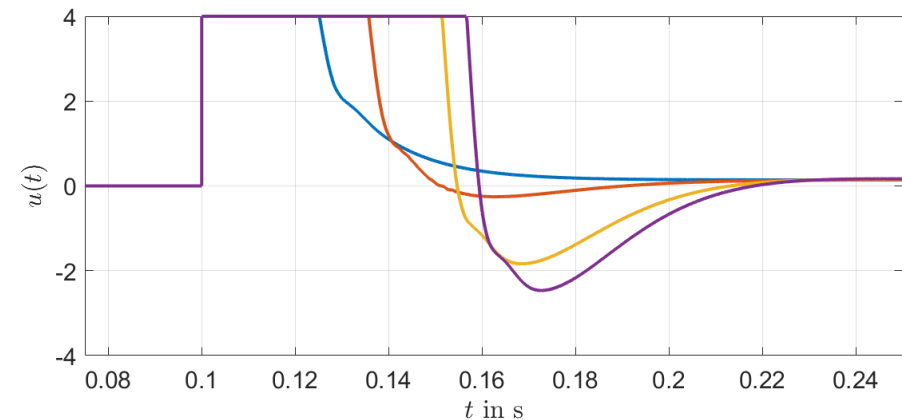
$$\frac{\tau_i s + 1}{\tau_i s} \quad \text{with} \quad \frac{\tau_i s + 1}{\tau_i s + \frac{1}{\beta}} \quad (\text{reduce low-frequency gain})$$

The new open-loop transfer function does not have a pure integrator so the steady-state error is not guaranteed to be zero but it will be reduced by a factor of $\beta > 1$.

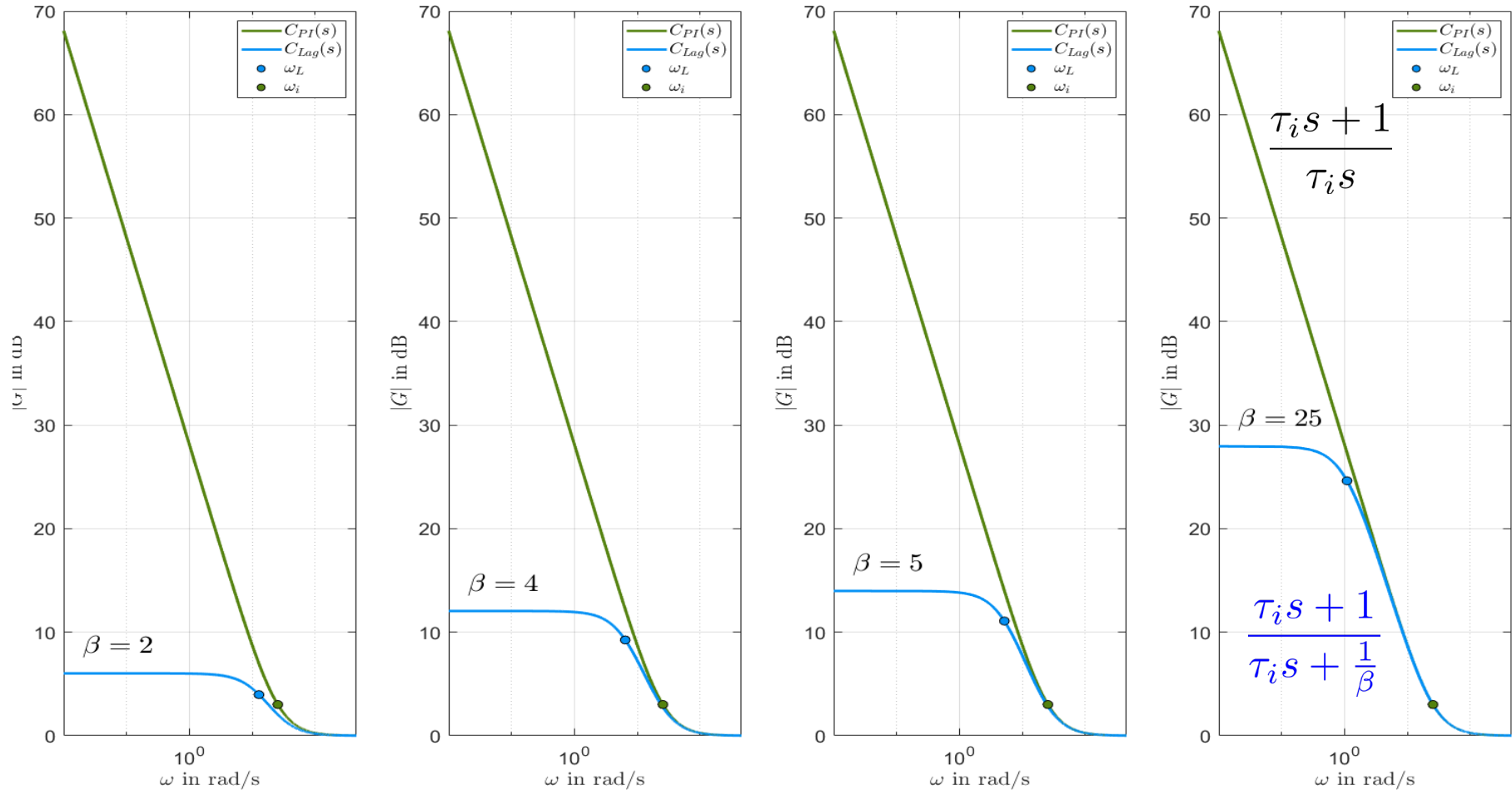
$$\text{P-Lead-Lag controller: } C(s) = K_P \frac{\tau_d s + 1}{\alpha \tau_d s + 1} \frac{\tau_i s + 1}{\tau_i s + \frac{1}{\beta}}$$



Theoretical exercise 3



Amplitude limitations on the system's input



Amplitude limitations on the system's input

Example (adapted from Exam 21)

Design a P-Lead-Lag controller for the system

$$G(s) = \frac{5000(s+7)(s+12)}{(s+4)(s+9)(s+17)(s+5)(s+70)}$$

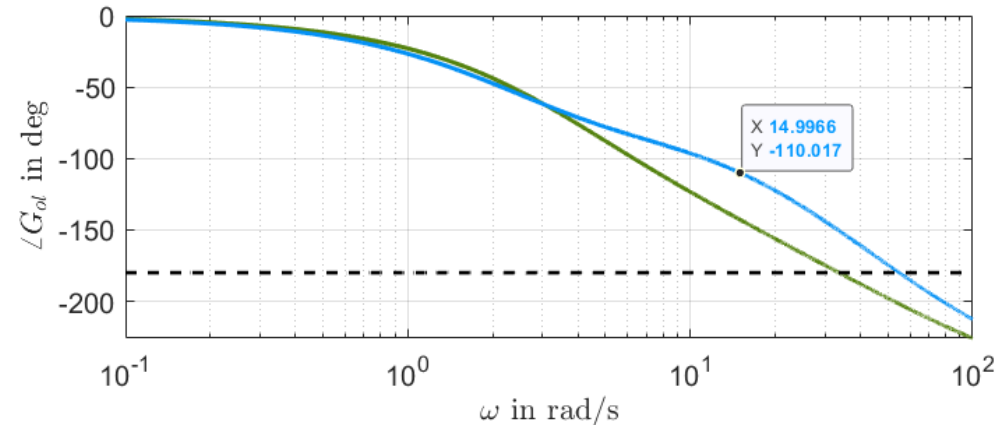
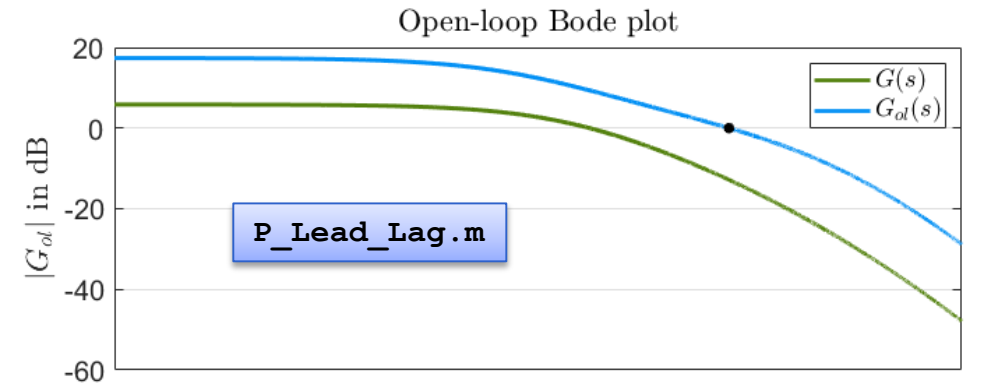
with phase margin $\gamma_M = 70^\circ$, $\alpha = 0.2$, $\beta = 2$ and $N_i = 3$.

At the new crossover frequency we have

$$-\pi + \gamma_M = \angle G(j\omega_c) + \arctan\left(\frac{3(1-2)}{1+2 \cdot 3^2}\right) + \arcsin\left(\frac{1-0.2}{1+0.2}\right)$$

$$\Rightarrow \omega_c \approx 15 \text{ rad/s} \Rightarrow \tau_i = \frac{N_i}{\omega_c} = 0.2, \tau_d = \frac{1}{\omega_c \sqrt{\alpha}} = 0.1491$$

$$K_P = \frac{1}{|C_{PI}(s)C_D(s)G(s)|}\bigg|_{s=j\omega_c} = 1.88$$



Amplitude limitations on the system's input

Example (adapted from Exam 21)

Design a P-Lead-Lag controller for the system

$$G(s) = \frac{5000(s+7)(s+12)}{(s+4)(s+9)(s+17)(s+5)(s+70)}$$

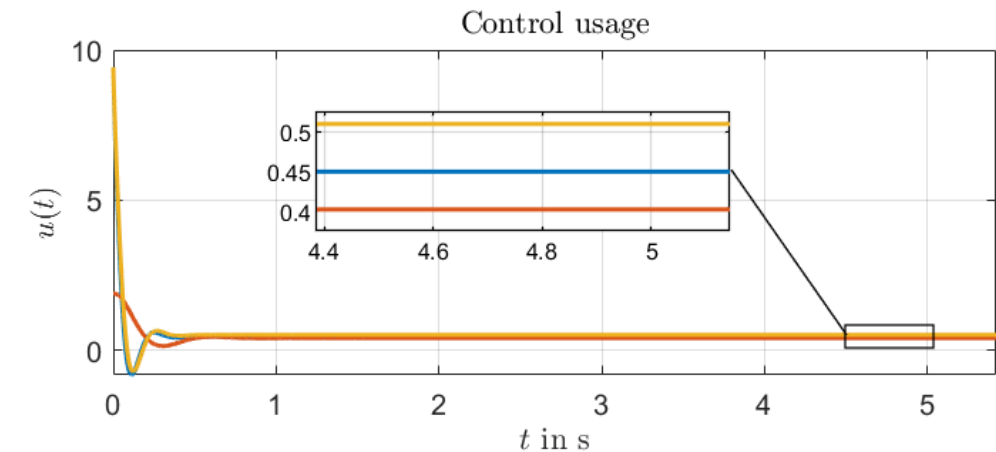
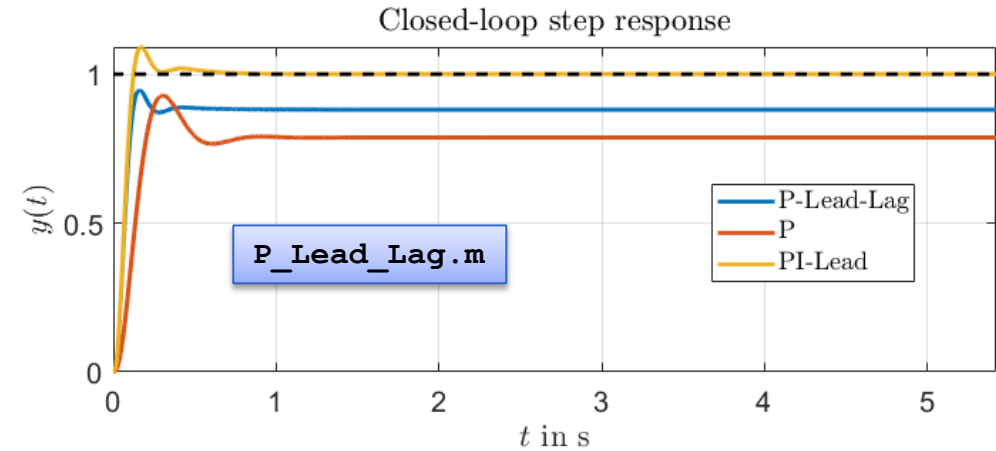
with phase margin $\gamma_M = 70^\circ$, $\alpha = 0.2$, $\beta = 2$ and $N_i = 3$.

At the new crossover frequency we have

$$-\pi + \gamma_M = \angle G(j\omega_c) - \arctan\left(\frac{3(1-2)}{1+2 \cdot 3^2}\right) + \arcsin\left(\frac{1-0.2}{1+0.2}\right)$$

$$\Rightarrow \omega_c \approx 15 \text{ rad/s} \Rightarrow \tau_i = \frac{N_i}{\omega_c} = 0.2, \tau_d = \frac{1}{\omega_c \sqrt{\alpha}} = 0.1491$$

$$K_P = \frac{1}{|C_{PI}(s)C_D(s)G(s)|}\bigg|_{s=j\omega_c} = 1.88$$



Amplitude limitations on the system's input

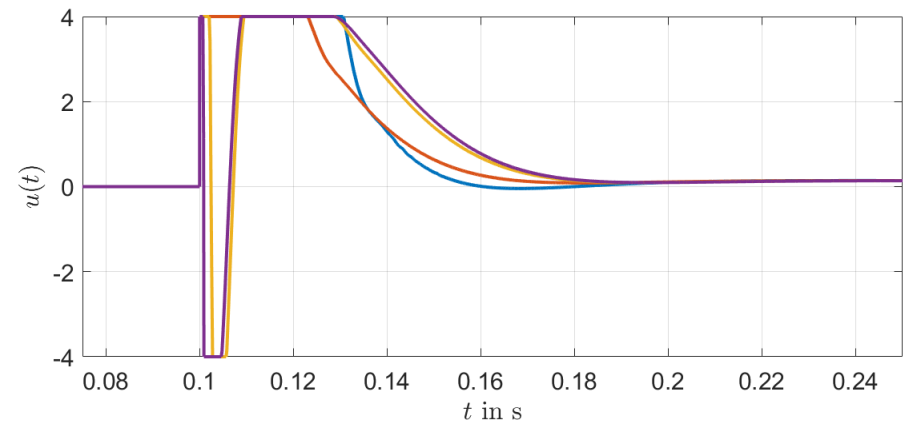
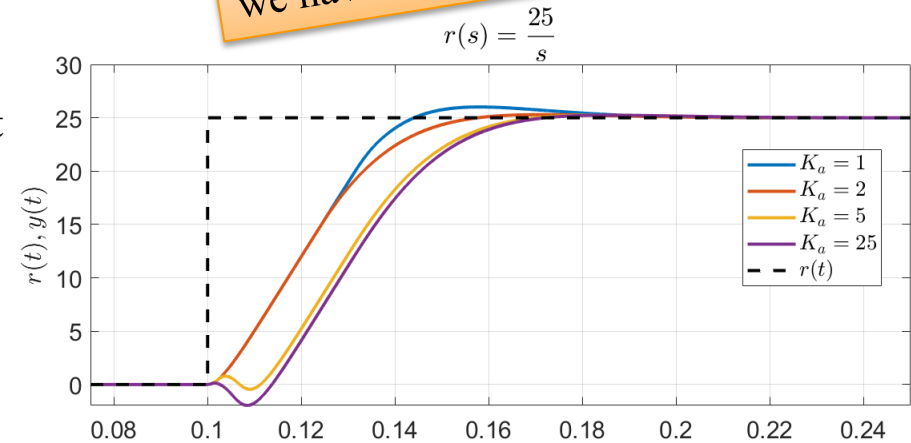
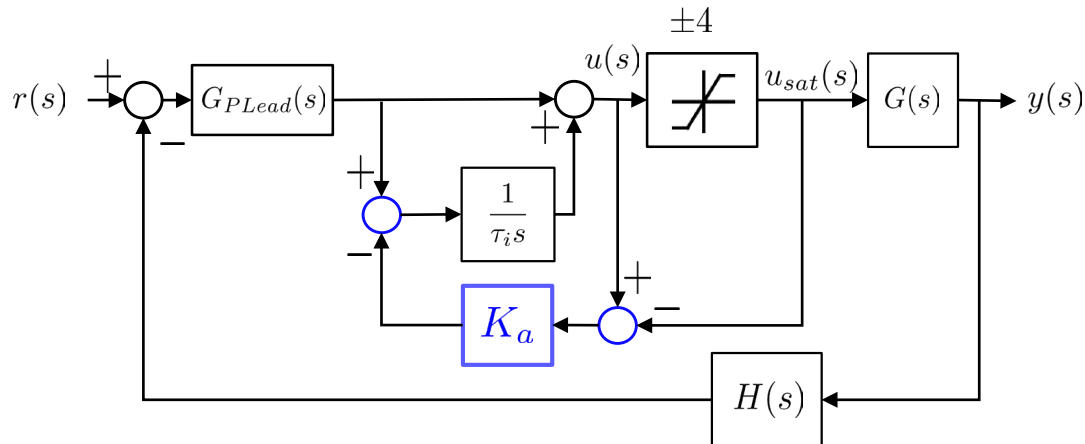
Mini quiz: Why do we have undershoots?

Integrator wind-up mitigation strategies

4. Anti-windup: Dynamically reduce the input to the integrator based on the difference between intended (not saturated) and actual (saturated) input to the system.

If the input is not saturated ($u(t) = u_{sat}(t)$), then the integrator is not limited.

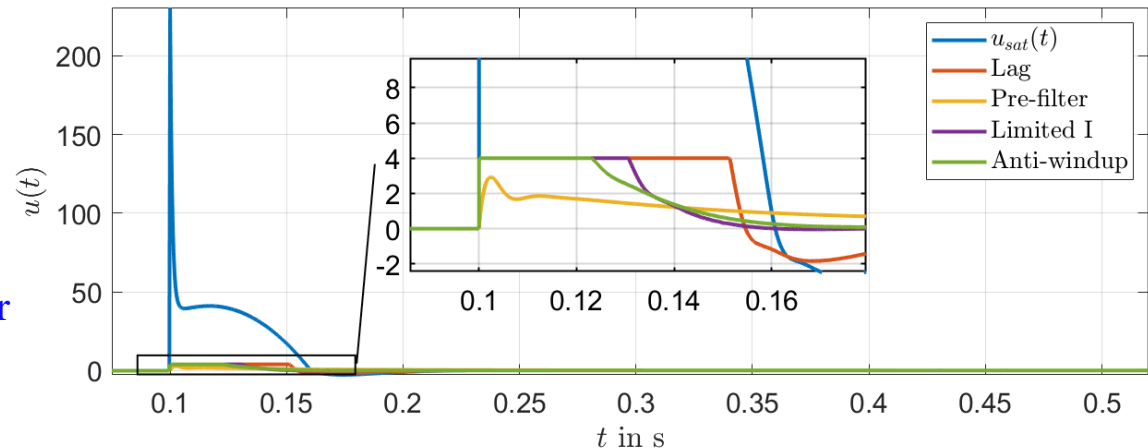
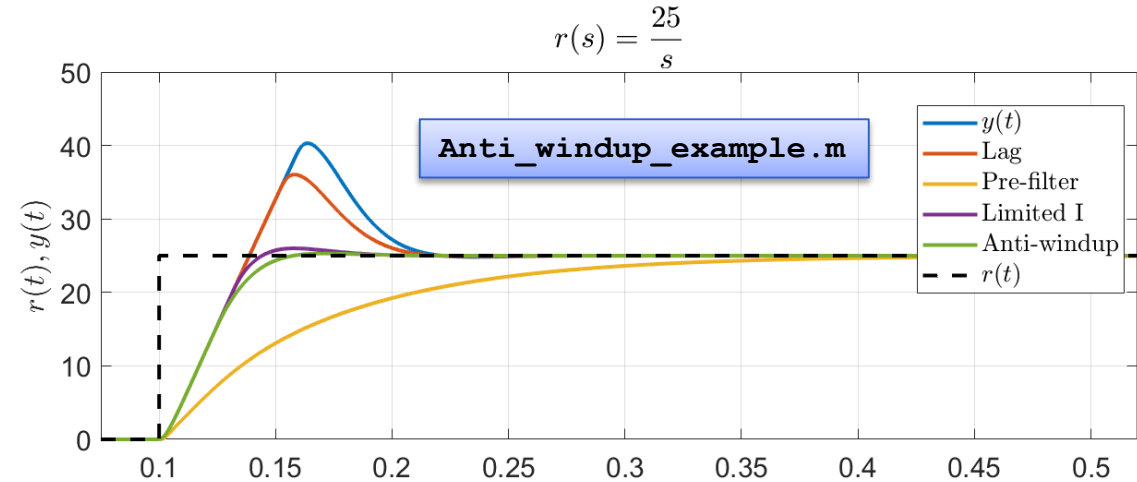
K_a is the factor by which the integrator input is reduced.



Amplitude limitations on the system's input

Integrator wind-up mitigation strategies

1. Limit the output of the integrator
 - Simplest solution
 - Integrator limit close to the input saturation
2. Insert a reference pre-filter
 - Simple solution
 - Makes the system slow even for small signals
3. Lag part instead of PI (P-Lead-Lag)
 - Reduces low frequency gains
 - Non-zero steady state error (reduced by a factor of β)
4. Anti-windup
 - Dynamic reduction of integrator
 - K_a should not be too large to avoid undershoots



Amplitude limitations on the system's input

Comprehension questions

- Is a input amplitude-limited (saturated) system nonlinear?
 - Yes and this causes our initial design to possibly fail.
- How can we find the transfer function of a input-saturated system?
 - From laws of physics plus linearisation (white box modelling).
 - From measurements (identification) but we need to use small inputs such that the input amplitude limitation (saturation) is not activated.
- Will a closed-loop system with a PI always be stable if the input amplitude is limited (saturation)?
 - No, stability is not guaranteed, especially for large reference signals, and we need to consider anti-windup solutions.
- A pre-filter can be a solution. Are there any disadvantages?
 - A pre-filter will make the closed-loop system slow, even for small signals.
- Are systems with input saturation common?
 - Yes, almost all systems have actuator saturation blocks for protecting against wear and tear as well as for safety reasons (e.g. limited torque delivered by a robotic manipulator).

Today you learned

- What is an input rate limitation in systems and how to deal with it
 1. Place the crossover frequency close to the dominant pole
- What is an input amplitude limitation in systems and how to deal with it (anti-wind up strategies)
 1. Integrator saturation
 2. Use of pre-filters
 3. P-Lead-Lag control design
 4. Anti-windup (feedback) design

Next lecture 12

Disturbances and Sensitivity transfer functions

- Find the transfer function from a disturbance to the output
- Calculate the steady-state error due to a disturbance
- Explain the sensitivity of system output and control input to disturbances

Today's Exercises

Group work: Continue on REGBOT balance (mandatory assignment)

Home alone: Continue on the theoretical exercises LCD1: 1, 2, 3, 4, 5, 6, 7, 9